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Decision focused online learning for real time energy aware scheduling of interconnected data centers with photovoltaic generation and battery storage

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ABSTRACT

Data centers require intraday scheduling that balances energy costs with renewable consumption. However, existing methods struggle with multi-source uncertainties in forecasting electricity prices, rooftop PV generation, and task arrivals. We propose a decision-focused online learning optimal scheduling method, which embeds the downstream optimization into forecasting model training, using decision loss rather than prediction error as the objective so forecasts directly serve scheduling quality. The method integrates forecasts of electricity prices, PV generation, and task arrivals: day-ahead training on historical samples captures long-term trends, while intraday rolling execution with delayed parameter updates — executing only the first action per step and updating after true values are observed — fully exploits real-time information. A regularized surrogate loss handles the non-differentiability of storage charging–discharging complementarity constraints during backpropagation. Experiments on real operational data from a data center cluster (June–December 2025) show monthly energy cost reductions of 16.5%–32.4% over conventional predict-then-decide methods, and cumulative PV consumption improving from 64.1% to 78.3% over seven months, confirming that decision-focused online learning effectively leverages real-time measurements to continuously improve both scheduling economy and renewable energy utilization.

Keywords: Online learning, decision-focused learning, computation-power coordination, energy storage optimization, renewable energy forecasting.

List of Main Symbols

$\mathcal{U}$	set of data center nodes, with node indices u, v
$\mathcal{T}$	set of dispatch periods, with period index t
$\mathcal{E}$	set of directed communication links between nodes
$\mathcal{D}$	set of task buffering tiers, with tier index d
i	day-ahead training sample index
H	length of the intraday rolling forecasting and delayed training window
$c, \hat{c}$	true and forecast exogenous parameter vectors
x, θ	input feature vector and forecasting model parameters
$J(w, c)$	operating cost (AUD) of decision w under parameters c
w, w^*	scheduling decision vector and its optimal solution
$\mathcal{W}$	overall scheduling feasible region
$\mathcal{W}^p, \mathcal{W}^s, \mathcal{W}^{dc}, \mathcal{W}^R$	feasible regions of power balance, energy storage, data center tasks, and renewable generation
$\mathcal{W}^{s,rel}, \mathcal{W}^{s,reg}(\epsilon)$	storage convex relaxed and regularized feasible regions
$\mathcal{I}_t$	operational observation information available at time t
$\lambda_{t,u}, \hat{\lambda}_{t,u}$	true and forecast task arrival volume (tasks/timeslot)
$P_{t,u}^R, \hat{P}_{t,u}^R$	true and forecast available renewable power (MW)
$\pi_{t,u}^{buy}, \pi_{t,u}^{sell}$	electricity purchase and sale prices (AUD/MWh)
$P_{t,u}^{buy}, P_{t,u}^{sell}$	power purchased from and sold to the grid (MW)
$P_{t,u}^{DC}$	total electrical load of a data center (MW)
$w_{t,u}^c, w_{t,u}^d$	storage charging and discharging power (MW)
$E_{t,u}$	storage energy state (MWh)
$w_{t,u}^{R,use}, w_{t,u}^{R,cur}$	renewable consumption and curtailment power (MW)
$q_{t,u}^{(d)}, r_{t,u}^{(d)}$	task volume before and after processing in tier d (tasks/timeslot)
$p_{t,u}^{(d)}$	task processing volume in tier d (tasks/timeslot)
$y_{t,u \rightarrow v}^{(d)}$	task volume migrated from node u to node v in tier d (tasks/timeslot)
$\alpha_{t,u}, \beta_{t,u}$	IT equipment power regression coefficients (MW-timeslot/tasks), (MW)
η_c, η_d	storage charging and discharging efficiencies
γ^{ess}, γ^R	storage cycling and renewable curtailment penalty coefficients
Θ	feasible region of forecasting model parameters
Π_{Θ}	projection operator of the parameter vector onto the feasible region Θ
N	total number of day-ahead training samples
$\mathcal{S}_{tr}, \mathcal{S}_{val}, \mathcal{S}_{te}$	training, validation, and test sets
Δt	dispatch period length (h)
D	maximum task buffering tier
s	data volume coefficient per unit task (Gbps-timeslot/tasks)
$\bar{B}_{t,uv}$	communication link bandwidth upper bound from node u to node v (Gbps)
$\mu_{max,u}$	maximum single-period task processing capacity of node u (tasks/timeslot)
$\bar{w}_u^c, \bar{w}_u^d$	storage charging and discharging power upper bounds (MW)
$\bar{E}_u, \underline{E}_u$	storage energy upper and lower bounds (MWh)
E_t^{phys}	physical storage energy state after execution (MWh)
η_k, η_t	day-ahead and intraday learning rates
α_k, α_t	quadratic smoothing coefficients of the surrogate objective
ϵ	complementarity constraint regularization parameter
$\Phi_{t,u}$	piecewise smooth regularization function for the storage complementarity constraint
ρ	constraint violation penalty coefficient
Δ	SPSA Rademacher random perturbation direction
c_k, c_t	SPSA perturbation radii
$\hat{\nabla}, \bar{\nabla}$	analytic gradient and SPSA zeroth-order gradient estimate
ϑ_s	temporary oracle parameter of the delayed training window
$\mathcal{L}_{DA}$	day-ahead decision-focused learning loss function
η	PV consumption rate

1 Introduction

Data centers are critical infrastructure for the digital economy, and their energy consumption continues to rise with the growing demand for cloud computing, artificial intelligence, and high-performance computing. Electricity costs, carbon emission constraints, and power supply reliability have become key operational bottlenecks for data centers¹⁻³. To reduce grid electricity purchase costs and improve green energy utilization, an increasing number of interconnected data centers are integrating distributed and rooftop PV generation, forming campus- or microgrid-level energy systems that can interact with the grid^{4,5}. However, renewable generation is volatile and intermittent; distributed generation alone cannot guarantee continuous power supply and economic operation of computing loads. Consequently, data centers typically further deploy electrochemical energy storage or repurpose backup battery resources for peak shaving, PV consumption, demand response, and reliability support^{6,7}. In this context, energy optimization for data centers equipped with renewable generation and energy storage is no longer a simple load reduction problem but a dynamic scheduling problem governed by the joint effects of electricity prices, PV generation, computing task arrivals, storage state of charge, and service delay constraints^{3,8}.

Energy storage systems (ESS) decouple energy production from consumption in the time dimension through flexible charging and discharging control, and this capability has made them a core flexibility resource for smoothing fluctuations and reducing operating costs in power systems with a high share of renewable generation⁹⁻¹¹; likewise, in data center energy systems equipped with storage, the charging–discharging strategy is deeply coupled with price signals, green power output, and the time sequence of computing tasks. On this basis, an energy management system (EMS) coordinates storage, distributed generation, and diverse loads to achieve multi-energy complementarity and economic–low-carbon multi-objective optimal dispatch under uncertainty^{12,13}. Furthermore, to cope with rapidly changing intraday uncertainties, some studies have introduced enhanced strategies such as deep-learning-driven forecasting and decision-making¹⁴, multi-agent game theory¹⁵, and day-ahead–intraday rolling corrections^{16,17} to improve the adaptability of dispatch frameworks. Modern large data centers are generally equipped with storage and can shift electricity usage in response to time-of-use prices and green power signals. However, storage operation in a data center is not an isolated decision: its charging–discharging strategy is tightly coupled with the spatiotemporal distribution of computing task loads, service level agreement (SLA) constraints, and task migration among multiple centers, which makes the coordinated scheduling of storage and computing resources more complex than in traditional power systems.

Conventional energy management mostly adopts a “predict-then-optimize” pipeline¹⁸: first forecast load, renewable output, or electricity prices, then feed the forecasts into the downstream scheduling model. This pipeline typically evaluates forecast quality using statistical metrics such as mean squared error (MSE), but smaller prediction error does not necessarily produce better scheduling outcomes, because the impact of forecast errors on decisions is often asymmetric, non-monotonic, and nonlinear^{19,20}. Decision-focused learning (DFL) provides a new methodological framework for bridging the gap between forecast accuracy and decision quality by embedding the optimization layer into the learning process and training the forecasting model with the downstream decision loss induced by predictions^{19,21}. In recent years, DFL has been applied in power systems to problems such as renewable energy trading and economic dispatch; related studies show that forecasting models trained toward decision loss can achieve economic outcomes superior to those that solely pursue statistical accuracy in several scheduling tasks^{22,23}.

Table 1 systematically compares recent power-system DFL studies from the dimensions of application scenario, decision training and execution mode, multi-source forecast coordination, spatiotemporal constraint coupling, intraday online learning, and training treatment of the storage complementarity constraint. Existing power-system DFL research has verified the basic idea that “forecasting serves decision-making,” but its forecasting objects are usually limited to a single energy variable such as load, wind power, or electricity price, and the decision tasks are mostly a single business link such as energy trading, unit commitment, or economic dispatch. Even when load and reserve are considered jointly, the main coupling relationship remains the power balance inside the power system, and task arrivals, cross-center migration, and service quality constraints are not incorporated into the same decision loss. Therefore, existing frameworks cannot express the cross-domain, cross-time transmission chain of “task prediction error–queue backlog–spatial migration–node load–storage operation,” and cannot make task arrivals, electricity prices, and PV generation jointly learn against the operating cost of interconnected data centers.

From the perspective of the decision execution mechanism, most of the methods in the table still rely on offline training with historical samples, forming a trading or scheduling plan in one shot within a given forecast window; those studies that include day-ahead and real-time corrections essentially do not perform delayed updates of the forecasting model based on the true information revealed after each first-action execution. Although general Online DFL provides a theoretical mechanism for parameter updates under delayed feedback³⁵, it does not handle the controlled state composed of storage state of charge, task queues, and migration results, nor does it provide an implementable scheduling closed loop of “rolling optimization–first-action execution–true information revelation–model update” under multi-center physical constraints. Consequently, existing offline DFL models cannot be directly migrated into the intraday online scheduler required here, and general online learning frameworks cannot directly replace the computational–energy joint state modeling and rolling execution mechanism of this paper.

In summary, although the above studies have advanced the application of DFL in power systems, their operational paradigm differs from that of intraday energy optimization for interconnected data centers, which presents four challenges: coupling of multi-source forecasting objects, spatial constraint coupling, intraday online closed-loop operation, and cross-temporal state propagation.

1) Multi-source forecasting object coupling: Existing DFL applications in power systems mostly focus on a single forecasting object or a single scheduling stage, such as renewable energy trading, economic dispatch, or electricity-price-driven energy storage arbitrage^{22,23,34}. In contrast, interconnected data centers with renewable generation must simultaneously handle multi-source uncertainties including regionally differentiated electricity prices, distributed PV generation, and computing task arrivals, and jointly map them to cross-center task migration, storage charging–discharging, and renewable consumption decisions.

2) Spatial constraint coupling: Existing DFL studies seldom characterize the spatial coupling constraints within clusters of interconnected data centers. Electricity prices and PV generation vary across regions; task migration is constrained by communication bandwidth, processing capacity, and service delay limits; and queue backlogs propagate continuously across nodes and time periods. As a result, forecast errors not only affect individual device actions but also alter the allocation of tasks among different data centers and the subsequent scheduling states. Single-center or single-energy-storage-arbitrage models cannot directly capture these effects.

3) Intraday online closed-loop operation: Typical DFL training mostly adopts offline samples or day-ahead batch training assumptions, which cannot directly reflect the closed-loop characteristics of intraday rolling operation—namely, gradual information revelation, continuous state transition, and immediate decision execution^{21,35}. A decision-focused electricity price forecasting method proposed for energy storage arbitrage³⁴ has demonstrated that the relationship between price forecast error and storage arbitrage profit is not simply consistent, and designed a hybrid loss with stochastic gradient descent training. However, that study mainly targets storage arbitrage model training within a given forecast horizon, without considering the importance of intraday online decision-making—that is, it does not characterize the actual operational process in which only the first action of the rolling optimization is executed at each time step and model parameters are subsequently updated based on newly revealed electricity price and renewable generation information.

4) Cross-temporal state propagation: For interconnected data centers, both the storage state of charge and task queue backlogs propagate across time steps, and task migration further alters the load distribution of subsequent nodes. If the intraday online decision and delayed update mechanism is ignored, the forecasting model obtained from day-ahead training cannot adapt in a timely manner to short-term disturbances such as price spikes, PV shading, and task surges^{35,36}.

5) Embedding DFL into data center scheduling with energy storage also faces numerical stability challenges. Storage charging and discharging power cannot be simultaneously positive; the complementarity condition $P_t^c P_t^d = 0$ forms a mathematical program with complementarity constraints (MPCC)^{37,38}. The MPCC feasible region is non-convex and non-smooth, often violating classical constraint qualifications, and the KKT convergence interpretation of general-purpose NLP solvers does not directly hold^{39,40}. Existing exact convex relaxation, penalty function, and smoothing methods^{37,39,41–43} each have their applicable conditions; when coupled with round-by-round online learning, rolling execution, and multi-source forecasting, gradient singularities, training oscillations, and computational burden are further amplified.

The problem addressed in this paper is: for multiple interconnected data centers with distributed PV, energy storage, and task migration constraints, construct a DFL online scheduling method that can simultaneously leverage day-ahead historical samples and intraday measurement information, so that the forecasting model directly serves rolling energy scheduling decisions. The main contributions of this paper are as follows.

First, we propose a multi-source predictive decision-focused online scheduling framework for interconnected data centers, which jointly maps regional electricity prices, distributed PV output, and computing task arrivals into an energy cost objective. Compared with the single-objective or single-stage studies listed in Table 1, this paper enables multi-source uncertainties to jointly influence cross-center task migration, storage charging–discharging, and renewable consumption decisions.

Second, we construct a two-stage method combining day-ahead training with intraday online rolling decision-making. The day-ahead stage uses historical samples to learn long-term trends, providing a prior model for intraday operation; in the intraday stage, only the first scheduling action is executed in each rolling window, and delayed training of the forecasting model is performed after true electricity prices, PV generation, and task arrivals are gradually revealed. This design implements the delayed-feedback idea of general Online DFL in an actual scheduling closed loop that includes storage state-of-charge and task queue state transfer.

Third, we construct a spatiotemporal task transfer model for interconnected data centers that can be embedded in the DFL training process, uniformly characterizing FIFO queue backlogs, SLA service delays, link bandwidth, and cross-center task migration. This model enables task arrival forecast errors to propagate through queue states and spatial allocation into subsequent energy decisions, covering a computational–energy coupling link not yet addressed by existing energy DFL studies.

Fourth, we design a hybrid surrogate loss and regularized online training method for the storage complementarity constraint,

Table 1. Systematic comparison of recent power-system DFL studies

Study	Application scenario	Decision training and execution mode	Multi-source forecast coordination	Spatiotemporal constraint coupling	Intraday online learning	Training treatment of storage complementarity constraint
Zeng et al. ²⁴	Unit commitment	Offline decision	Thermal unit ramping	Storage state transition	No	Integer variables
Jin et al. ¹⁵	Integrated energy system dispatch	Offline decision	Renewable generation, price	Storage state transition	No	Integer variables
Li et al. ¹⁶	Microgrid economic dispatch	Offline decision	Renewable generation, load	Storage and EV state transitions	No	Integer variables
Kalantari et al. ¹⁷	Virtual power plant trading	Day-ahead offline, intraday online decisions	Renewable generation, load	Storage and EV state transitions, network flow	No	Integer variables
Carrière et al. ²²	Renewable trading	Offline decision	Renewable generation	None	No	Not addressed
Lu et al. ²³	Power system economic dispatch	Offline decision	Load	None	No	Not addressed
Sang et al. ³⁴	Storage arbitrage	Offline decision	Electricity price	Storage state transition	No	Convex upper bound constructed by an auxiliary storage problem
Chen et al. ²⁵	Network-constrained unit commitment	Offline decision	Load	None	No	Not addressed
Zhang et al. ²⁶	Power trading	Day-ahead offline, intraday online decisions	Load	None	No	Not addressed
Garcia et al. ²⁷	Power trading	Offline decision	Load, reserve	None	No	Not addressed
Wang et al. ²⁸	Unit commitment	Indirect gradient DFL; day-ahead dispatch with real-time correction	Load	None	No	Not addressed
Han et al. ²⁹	Economic dispatch	Indirect gradient DFL; day-ahead stochastic economic dispatch	Load	None	No	Not addressed
Muñoz et al. ³⁰	Renewable trading	Indirect gradient-free DFL; offline trading decision	Wind power	None	No	Not addressed
Zhao et al. ³¹	Wind power bidding	Indirect gradient-free DFL; interval-forecast-based offline decision	Wind power interval	None	No	Not addressed
Stratigakos et al. ³²	Renewable trading	Direct gradient-free DFL; prescriptive-tree offline decision	Renewable output	None	No	Not addressed
Wahdany et al. ³³	Renewable trading	Indirect gradient DFL; end-to-end offline trading decision	Wind power	None	No	Not addressed
This paper	Economic operation of interconnected data centers	Day-ahead offline decision, intraday online decision with delayed training	Price, PV output, task arrivals	Task backlog, storage state transition	Yes	Regularized surrogate loss

addressing the non-differentiability and numerical instability of the non-simultaneous charging–discharging constraint in DFL updates. The method distinguishes the training surrogate layer from the actual execution layer and combines it with multi-round intraday delayed updates, improving the stability and feasibility of online parameter updates.

The remainder of this paper is organized as follows. Section 2 introduces the overall framework and basic rationale of decision-focused learning optimal scheduling. Section 3 establishes the detailed optimal scheduling model for interconnected data centers, covering data center topology, spatiotemporal task transfer, energy storage operation, and power balance constraints. Section 4 presents the solution approach combining day-ahead training and intraday online rolling decision-making, and discusses strategies for handling the energy storage complementarity constraint. Section 5 validates the effectiveness of the proposed method through simulation case studies and discusses the influence of parameters and algorithm components on the experimental process and results. Section 6 concludes the paper. Section 7 discusses the applicability boundaries of the modeling assumptions and outlines future work.

2 Optimal Scheduling Framework Based on Decision-Focused Learning

2.1 Day-Ahead Decision-Focused Learning Optimal Dispatch

Day-ahead decision-focused learning optimal dispatch (DA-DFLOD) provides a prior reference for intraday decision-focused online learning optimal dispatch (Online DFLOD). Its training method can be formulated as a bilevel problem:

$$\begin{aligned} \min_{\theta \in \Theta} \sum_{i=1}^N [J(w_i^*(\hat{c}_i), c_i) - J(w_i^*(c_i), c_i)] \\ \hat{c}_i = m_\theta(x_i) \\ w_i^*(\hat{c}_i) \in \operatorname{argmin}_{w_i \in \mathcal{W}_i} J(w_i, \hat{c}_i) \end{aligned} \quad (1)$$

where θ denotes the trainable parameters of the forecasting model, and Θ its feasible region; $\hat{c}_i = m_\theta(x_i)$ is the parameter vector output by the forecasting model for the i -th sample (e.g., electricity prices, renewable generation, etc.), and c_i the corresponding true parameter vector; w_i denotes the scheduling decision variables of the i -th sample (e.g., storage charging/discharging power, computing task allocation, etc.), and $\mathcal{W}_i$ their feasible region; $w_i^*(c_i)$ is the optimal decision of the i -th sample given the true parameters c_i , i.e., $\operatorname{argmin}_{w_i \in \mathcal{W}_i} J(w_i, c_i)$, and $w_i^*(\hat{c}_i)$ the optimal decision given the forecast parameters $\hat{c}_i$; $J(w, c)$ denotes the operational cost incurred by taking decision w under parameters c ; $\sum_{i=1}^N$ accumulates the decision regret over N training rounds, i.e., the difference between the forecast-induced decision cost and the optimal decision cost. The training process is illustrated in Fig. 1:

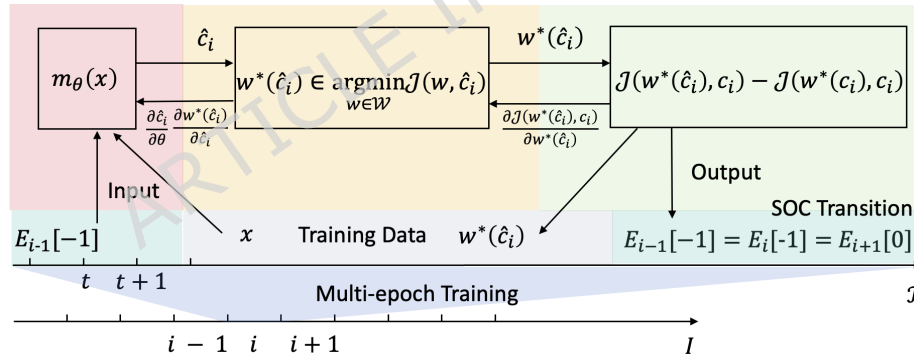


Figure 1. DA-DFLOD model training structure

The data center contains dispatchable energy storage, which involves energy state transition constraints. If long-horizon training data are fed in, the computational cost will be very high. To this end, the DA-DFLOD training structure adopts a parallel training approach and achieves decoupling by ensuring energy equality between the start and end of each training round, i.e., $E_{i,u,0} = E_{i,u,N_i}$ (N_i being the number of periods covered by the i -th training window); adjacent rounds satisfy $E_{i-1,u,N_{i-1}} = E_{i,u,0}$ and $E_{i,u,N_i} = E_{i+1,u,0}$. The training set of the day-ahead model typically spans a one-year horizon to capture seasonality.

In Fig. 1, arrows from left to right represent forward propagation, and arrows from right to left represent backward propagation. During backpropagation training, the gradient of the optimal value with respect to the forecast values is generally difficult to compute directly, typically because the decision variables may be discontinuous, non-differentiable, or zero-valued. The specific nature of the gradient depends on the form of the optimization model (the mathematical formulation of which will be presented in the next section). We suppress gradient singularities through regularization methods and complete the training with the aid of perturbation techniques. The day-ahead training procedure above is summarized as Algorithm 1.

Algorithm 1: Day-ahead decision-focused learning training

Input: training set $\mathcal{S}_{\text{tr}} \equiv \{(X_i, c_i)\}_{i=1}^{N_{\text{tr}}}$, validation set $\mathcal{S}_{\text{val}}$

Hyperparams: $\eta_k, \alpha_k, c_k, \varepsilon, \rho$, maximum rounds K

Initialize $\theta_0 \in \Theta$, compute and fix training-set standardization statistics

for $k = 0, 1, \dots, K-1$ do

for each training window $(X_i, c_i) \in \mathcal{S}_{\text{tr}}$ do

$$\hat{c}_i = m_{\theta_k}(X_i)$$

$$w_i^{*,\text{re}} \in \underset{w_i \in \mathcal{W}_i}{\operatorname{argmin}} J(w_i, \hat{c}_i)$$

$$\tilde{w}_i(\theta_k) \in \underset{w_i \in \mathcal{W}_i}{\operatorname{argmin}} \left\{ J(w_i, \hat{c}_i) + \frac{\alpha_k}{2} \|w_i\|_2^2 \right\}$$

$$L_i(\theta_k) = \frac{J(\tilde{w}_i(\theta_k), c_i) - J(w_i^*(c_i), c_i)}{s_J} + \rho v_i(\theta_k)$$

(2)

$$\Delta \sim \{-1, 1\}^p$$

$$\theta_k^+ = \Pi_{\Theta}(\theta_k + c_k \Delta), \quad \theta_k^- = \Pi_{\Theta}(\theta_k - c_k \Delta)$$

$$\hat{\nabla} L_i(\theta_k) \text{ by two-sided SPSA using } L_i(\theta_k^+), L_i(\theta_k^-)$$

end for

$$\hat{\nabla} L(\theta_k) = \frac{1}{|\mathcal{S}_{\text{tr}}|} \sum_{i \in \mathcal{S}_{\text{tr}}} \hat{\nabla} L_i(\theta_k)$$

$$\theta_{k+1} = \Pi_{\Theta}(\theta_k - \eta_k \hat{\nabla} L(\theta_k))$$

Evaluate decision loss, MSE and constraint violations on $\mathcal{S}_{\text{val}}$

if validation decision loss does not decrease, then early stop

end for

Output: θ_{DA} with the lowest validation decision loss

2.2 Intraday Decision-Focused Online Learning Optimal Dispatch

The decisions $w_i^*(\hat{c}_i)$ and training parameters θ_t from the DA-DFLOD model provide a reference baseline for Online DFLOD. The day-ahead model captures long-term characteristics and trends on a weekly cycle, while the Online DFLOD model corrects strategies in real time based on the latest observations, capturing short-term characteristics and trends. We define Online DFLOD as an online learning method: online learning refers to the learning paradigm in which the model incrementally updates its parameters each time a new observation arrives as sequential data gradually become available, rather than waiting for a complete dataset to be collected before one-shot training.

Online DFLOD scheduling consists of two steps: rolling execution and delayed training. In the rolling execution phase, at each time step t , the Online DFLOD model receives operational observation information $\mathcal{S}_t$ and the state transition variables from the previous time step—for energy storage this is E_t^{phys} , and for data centers with queue-based processing this is the task backlog (the figure only shows the storage state variable). Given the forecasting model parameters θ_t , the model forecasts a look-ahead window of length H from time t , generates future-window forecasts, and solves for the complete action sequence:

$$\hat{c}_{t:t+H-1} = m_{\theta_t}(\mathcal{S}_t)$$

$$w_{t:t+H-1}^*(\hat{c}_{t:t+H-1}) \in \underset{w \in \mathcal{W}}{\operatorname{argmin}} J(w, \hat{c}_{t:t+H-1})$$

(3)

From the complete action sequence $w_{t:t+H-1}^*(\hat{c}_{t:t+H-1})$ obtained by solving the full window, only the first action is taken as the actually executed action $w_t = w_{t,0}^*$, based on which the state transition is computed (for energy storage, E_{t+1}^{phys}) and passed to the next window, thus completing one round of rolling execution. In the delayed training phase, once period $t-1$ has elapsed, i.e., upon entering time t , the true electricity prices, renewable generation, and other information for period $t-1$, $\{c_{t-H}, c_{t-H+1}, \dots, c_{t-1}\}$, are revealed. Shifting forward by H time windows forms the online training set, and its forward and

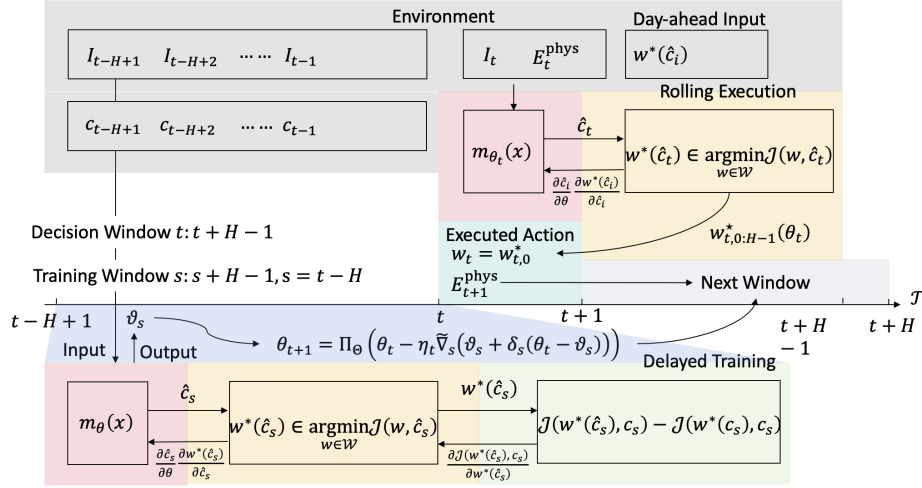


Figure 2. Rolling execution–delayed training structure of the Online DFLOD model

backward propagation is entirely consistent with that of the DA-DFLOD model training. Let the sliding training window with a one-step lag be $s : s + H - 1$, where $s = t - H$:

$$\begin{aligned}
 \vartheta_s &\in \argmin_{\vartheta \in \Theta} [J(w_{s:s+H-1}^*(\hat{c}_{s:s+H-1}), c_{s:s+H-1}) \\
 &\quad - J(w_{s:s+H-1}^*(c_{s:s+H-1}), c_{s:s+H-1})] \\
 \hat{c}_{s:s+H-1} &= m_{\vartheta}(\mathcal{J}_s) \\
 w_{s:s+H-1}^*(\vartheta) &\in \argmin_{w \in \mathcal{W}} J(w, \hat{c}_{s:s+H-1})
 \end{aligned} \tag{4}$$

Note that the output is ϑ_s ; the purpose of delayed training is not to fully replace the current θ_t , but to retain the trends the model has already learned from historical operational data, using stochastic gradient updates to avoid the problem of catastrophic forgetting. When the objective function is linear in the decisions, the update can be written analytically as

$$\begin{aligned}
 u_s &= \vartheta_s + \delta_s(\theta_t - \vartheta_s), \\
 \delta_s &\sim \text{Unif}([0, 1]), \\
 \tilde{\nabla}_s(u_s) &= \nabla_{\theta} \tilde{w}_{s:s+H-1}(u_s)^{\top} c_{s:s+H-1} \\
 \theta_{t+1} &= \Pi_{\Theta}(\theta_t - \eta_t \tilde{\nabla}_s(u_s)).
 \end{aligned} \tag{5}$$

For the generality of the framework, SPSA is used in the actual implementation. Here Π denotes the projection of θ onto the feasible region; $\tilde{w}$ denotes the solution of the regularized lower-level problem, used to avoid gradient singularities; and η_t is the dynamic gradient update step size. The detailed mathematical formulation will be elaborated in Section 4. Since the training window $s = t - H$ always lags behind the current decision window $t : t + H - 1$ by one complete window, and its endpoint $t - 1$ is one period behind the current time t , the procedure is referred to as delayed training. The intraday rolling execution and delayed training procedure above is summarized as Algorithm 2.

Algorithm 2: Intraday rolling execution and delayed training

Input: day-ahead parameters θ_{DA} , initial states $E_0^{\text{phys}}, r_0^{\text{phys}}$

Hyperparams: rolling window $H, \eta_t, \alpha_t, c_t, \varepsilon, \rho$

Initialize $\theta_0 = \theta_{DA}$

for $t = 0, 1, \dots$ do

Observe $\mathcal{J}_t, E_t^{\text{phys}}, r_t^{\text{phys}}$

$\hat{c}_{t:t+H-1} = m_{\theta_t}(\mathcal{J}_t)$

$w_{t:t+H-1}^{*,\text{re}} \in \underset{w \in \mathcal{W}_{t:t+H-1}}{\text{argmin}} J(w, \hat{c}_{t:t+H-1})$

Execute $w_t = w_{t,0}^{*,\text{re}}$

Update physical states $E_{t+1}^{\text{phys}}, r_{t+1}^{\text{phys}}$

Reveal realized c_{t-1} and append it to history

if $t \geq H$ then

$s = t - H$, construct delayed training window $s : s + H - 1$

$$\tilde{w}_s(\theta_t) \in \underset{w \in \mathcal{W}_{s:s+H-1}}{\text{argmin}} \left\{ J(w, m_{\theta_t}(\mathcal{J}_s)) + \frac{\alpha_t}{2} \|w\|_2^2 \right\} \quad (6)$$

$$L_s^{\text{ID}}(\theta_t) = \frac{J(\tilde{w}_s(\theta_t), c_{s:s+H-1}) - J(w_s^*(c_{s:s+H-1}), c_{s:s+H-1})}{s_J}$$

$+ \rho v_s(\theta_t)$

$\Delta \sim \{-1, 1\}^p$

$\hat{\nabla} L_s^{\text{ID}}(\theta_t)$ by two-sided SPSA

$\vartheta_s = \Pi_{\Theta} \left(\theta_t - \eta_t \hat{\nabla} L_s^{\text{ID}}(\theta_t) \right)$

$u_s = \vartheta_s + \delta_s(\theta_t - \vartheta_s), \quad \delta_s \sim \text{Unif}[0, 1]$

$\theta_{t+1} = \Pi_{\Theta} \left(\theta_t - \eta_t \hat{\nabla} L_s^{\text{ID}}(u_s) \right)$

else

$\theta_{t+1} = \theta_t$

end if

Record cost, regret, MSE and constraint violations

end for

It should be noted that Gurobi MIP is used for scheduling in the execution phase, and the quadratic regularization smoothing term is used in the training phase.

3 Optimal Scheduling Model for Interconnected Data Centers Based on Decision-Focused Learning

3.1 Data Center Topology

The power supply structure of a data center with renewable generation is shown in Fig. 3, where diesel generators and cooling equipment are omitted. $P_{t,u}^{\text{IT},(1)}$ and $P_{t,u}^{\text{IT},(2)}$ denote the IT equipment power consumption of the u -th data center in period t , respectively. Data centers can regulate server power consumption by shifting computing tasks across time and space, which is then converted to the overall electrical load of the data center via the PUE factor; detailed modeling is provided in the next section. An integrated storage–backup electrochemical battery is connected to the static transfer switch (STS, Static Transfer Switch) and solid-state transformer (SST, Solid-State Transformer); when a grid fault occurs, the STS switches the power supply to the storage and diesel generator. Distributed PV is connected to the low-voltage side.

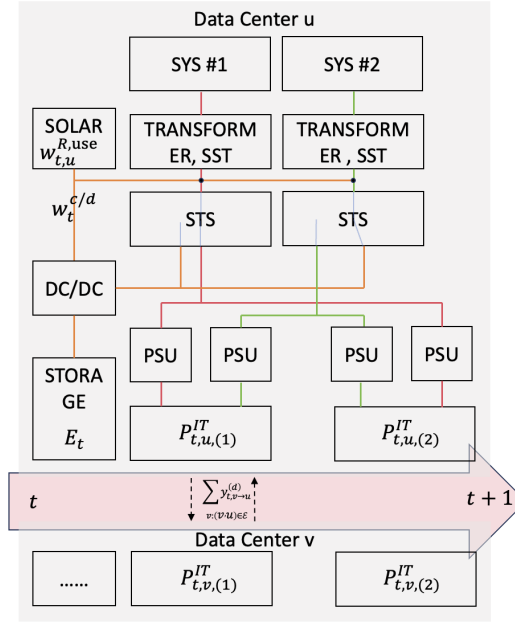


Figure 3. Data center power supply topology

3.2 Objective Function

This paper takes the minimization of the comprehensive energy cost of multiple data centers operated by a computing service provider as the scheduling objective. For the true outcomes $c = \{\pi_{t,u}^{\text{buy}}, \pi_{t,u}^{\text{sell}}, \lambda_{t,u}, P_{t,u}^R\}_{t \in \mathcal{T}, u \in \mathcal{U}}$, or the forecast outcomes $\hat{c} = \{\hat{\pi}_{t,u}^{\text{buy}}, \hat{\pi}_{t,u}^{\text{sell}}, \hat{\lambda}_{t,u}, \hat{P}_{t,u}^R\}_{t \in \mathcal{T}, u \in \mathcal{U}}$, the decision variables for energy storage, data centers, and renewable generation are denoted as $w = \{w^s, w^{dc}, w^R, P^{\text{buy}}, P^{\text{sell}}\}$, where $P_{t,u}^{\text{buy}}$ and $P_{t,u}^{\text{sell}}$ are the power purchased from and sold to the grid by the u -th data center in period t , respectively. The scheduling objective function is defined as

$$\min_{w \in \mathcal{W}} J(w, c) = \sum_{t \in \mathcal{T}} \sum_{u \in \mathcal{U}} \left[\pi_{t,u}^{\text{buy}} P_{t,u}^{\text{buy}} - \pi_{t,u}^{\text{sell}} P_{t,u}^{\text{sell}} + \gamma^{\text{ess}} \left(w_{t,u}^c + w_{t,u}^d \right) + \gamma^R w_{t,u}^{R, \text{cur}} \right] \Delta t, \quad (7a)$$

$$P_{t,u}^{\text{buy}} - P_{t,u}^{\text{sell}} = P_{t,u}^{\text{DC}} (\lambda_{t,u}) + w_{t,u}^c - w_{t,u}^d - w_{t,u}^{R, \text{use}}, \quad \forall t \in \mathcal{T}, u \in \mathcal{U}, \quad (7b)$$

$$P_{t,u}^{\text{buy}} \geq 0, \quad P_{t,u}^{\text{sell}} \geq 0, \quad \forall t \in \mathcal{T}, u \in \mathcal{U}. \quad (7c)$$

The above must hold for all $t \in \mathcal{T}$, $u \in \mathcal{U}$. Equations (7b)–(7c) constitute the power balance feasible region, denoted as $\mathcal{W}^p$. In (7a), the first term is the cost of purchasing electricity from the market, and the second term is the revenue from selling electricity back to the grid; if the operating scenario does not permit reverse power flow, set $P_{t,u}^{\text{sell}} = 0$. The third term is the cycling cost of energy storage charging and discharging; γ^{ess} is used to represent battery life degradation and the equivalent operating cost of frequent charging/discharging, preventing the model from generating meaningless charging/discharging actions under small price spreads. The fourth term is the penalty for renewable curtailment; γ^R is used to prioritize local consumption of renewable energy when economic costs are similar. When curtailment incurs no explicit settlement cost, a small positive value can be used, making it a secondary objective. Equation (7b) is the system grid-connected power balance: data center load and storage charging increase the electricity purchase requirement, while storage discharging and renewable consumption offset the purchase requirement. It should be noted that the interconnected data centers considered in this paper belong to the same computing service operator, and the inter-node migration traffic is carried by the operator's own backbone network without incurring market-based public network usage charges; the link bandwidth can be regarded as an already-invested fixed resource. Therefore, this paper only uses the bandwidth upper bound of (13b) to constrain migration feasibility and does not explicitly include migration cost in the objective function.

This objective function maps electricity price forecasts, computing task scheduling, renewable consumption, and storage operation into a single settleable energy cost. Electricity price forecasts for different regions influence the value of low-price charging, high-price discharging, and spatiotemporal task shifting through $\hat{\pi}_{t,u}^{\text{buy}}$ and $\hat{\pi}_{t,u}^{\text{sell}}$, computing task arrival volume

$\lambda_{t,u}$ affects the electrical load $P_{t,u}^{DC}$ of data centers at different nodes; available renewable power $P_{t,u}^R$ limits the consumable power $w_{t,u}^{R,use}$ and influences curtailment $w_{t,u}^{R,cur}$; storage decisions alter the net purchase curve through $w_{t,u}^c$ and $w_{t,u}^d$. Therefore, minimizing J can not only directly reduce the electricity expenditure of data centers but also provide a training signal for decision-focused learning that is consistent with actual operational profits, prompting the forecasting model to prioritize learning those error directions that would alter the scheduling cost, rather than merely pursuing statistical forecast accuracy.

3.3 Constraints

3.3.1 Energy Storage Model Constraints

This paper considers an integrated storage–backup electrochemical energy storage system, where energy losses during charging and discharging are non-negligible. The storage feasible region $\mathcal{W}^s$ includes power constraints, energy constraints (including state transition constraints), and complementarity constraints, for all $t \in \mathcal{T}$, $u \in \mathcal{U}$:

$$\begin{aligned}
 w_{t,u}^c &\leq \bar{w}_u^c \\
 -w_{t,u}^c &\leq -\underline{w}_u^c \\
 w_{t,u}^d &\leq \bar{w}_u^d \\
 -w_{t,u}^d &\leq -\underline{w}_u^d \\
 E_{0,u} + \sum_{\tau=0}^t (\eta_c w_{\tau,u}^c - \eta_d^{-1} w_{\tau,u}^d) \Delta t &\leq \bar{E}_u \\
 -E_{0,u} - \sum_{\tau=0}^t (\eta_c w_{\tau,u}^c - \eta_d^{-1} w_{\tau,u}^d) \Delta t &\leq -\underline{E}_u \\
 w_{t,u}^c w_{t,u}^d &= 0
 \end{aligned} \tag{8}$$

The storage charging and discharging decision variables are denoted as $w^s = \{w^c, w^d\}$. The complementarity constraint $w_{t,u}^c w_{t,u}^d = 0$ makes the lower-level problem non-convex. To facilitate convex optimization and the construction of the DFL surrogate decision layer, under certain conditions this constraint can be removed while retaining the power bounds, energy recursion, and state-of-charge bounds, yielding the storage convex relaxed feasible region:

$$\mathcal{W}^{s,rel} := \mathcal{W}^s \setminus \{w_{t,u}^c w_{t,u}^d = 0, \forall t \in \mathcal{T}, u \in \mathcal{U}\} \tag{9}$$

An exact convex relaxation means that the relaxed problem has the same optimal value as the original problem, and its optimal solution still satisfies non-simultaneous charging and discharging. Reference⁴¹ shows from the KKT conditions that when simultaneous charging and discharging only increases the marginal loss or operating cost without bringing additional energy revenue, the complementarity constraint can be removed exactly.

Combined with the objective function of this paper, the following sufficient condition can be used: the lower bounds of charging and discharging are zero, $0 < \eta_c \eta_d < 1$, and the grid permits purchase or reverse power flow according to (7b). Let $\tilde{\pi}_{t,u}^{buy}$ and $\tilde{\pi}_{t,u}^{sell}$ be the purchase and sale prices used by the current lower-level problem; if for any t, u ,

$$\gamma^{ess}(1 + \eta_c \eta_d) + (1 - \eta_c \eta_d) \min\{\tilde{\pi}_{t,u}^{buy}, \tilde{\pi}_{t,u}^{sell}\} > 0, \quad \forall t \in \mathcal{T}, u \in \mathcal{U}, \tag{10}$$

then (9) is an exact relaxation. Indeed, if an optimal solution charges and discharges simultaneously, both can be reduced proportionally so that $\eta_c w_{t,u}^c - \eta_d^{-1} w_{t,u}^d$ remains unchanged, thereby leaving the state of charge unchanged; the net load and cycling cost both decrease, contradicting optimality. In particular, when the purchase and sale prices are non-negative and $\gamma^{ess} > 0$, this condition holds automatically. Since the forecasting model in this paper allows negative electricity prices, condition (10) should be verified with forecast prices in each training or rolling window. In particular, when reverse power flow is prohibited ($P_{t,u}^{sell} = 0$), the grid-connected net exchange power is always non-negative and the relevant marginal price is $\tilde{\pi}_{t,u}^{buy}$; if the purchase price is non-negative and $\gamma^{ess} > 0$, condition (10) still holds automatically and the convex relaxation remains exact. If this sufficient condition no longer applies, the regularized complementarity feasible region proposed in Section 4 should be adopted, and the constraint violation $\max_{t \in \mathcal{T}, u \in \mathcal{U}} w_{t,u}^c w_{t,u}^d$ should be reported.

During multi-round day-ahead DA-DFLOD and intraday rolling decision-making, the storage state variable $E_{t,u} = E_{0,u} + \sum_{\tau=0}^t (\eta_c w_{\tau,u}^c - \eta_d^{-1} w_{\tau,u}^d) \Delta t$ must be passed between rounds.

3.3.2 Task Backlog and Spatiotemporal Transfer Constraints for Interconnected Data Centers

This section formulates cross-data-center task migration, queue backlog, and service delay constraints as a linear scheduling feasible region that can be embedded in decision-focused learning. The key to this modeling is to enable the task arrival forecast $\hat{\lambda}_{t,u}$ to directly influence the IT load, storage actions, and renewable consumption cost of each node through the lower-level scheduling model, thereby providing DFL with a training signal consistent with the energy scheduling profit. At the same time, the FIFO queue state $r_{t,u}^{(d)}$ can be passed between intraday rolling windows, and the cycle constraint can be decoupled between day-ahead training samples, allowing the model to characterize multi-node spatial coupling while controlling the solution scale and numerical burden across multiple rounds of DFL training.

Data center energy consumption mainly consists of IT equipment and cooling systems. A univariate regression server power model⁸ is adopted, assuming homogeneous servers and tasks³. Let $\alpha_{t,u}, \beta_{t,u}$ be the power regression coefficients, in units of (MW·timeslot/tasks); the total data center power consumption is determined by the server task volume:

$$p_{t,u}^{DC} = \alpha_{t,u} \sum_{d \in \mathcal{D}} p_{t,u}^{(d)} + \beta_{t,u}, \quad \forall t \in \mathcal{T}, u \in \mathcal{U} \quad (11)$$

In all equations, $u, v \in \mathcal{U}$, where $\mathcal{U}$ is the set of computing nodes; $t \in \mathcal{T}$, $\mathcal{T} = \{0, \dots, T-1\}$ is the set of time periods; $\mathcal{E} \subseteq \mathcal{U} \times \mathcal{U}$ is the set of directed communication links between computing nodes; $\mathcal{D} = \{0, \dots, D\}$, where d is the task buffering tier index and D the maximum buffering tier.

The spatiotemporal transfer characteristics of tasks in interconnected data centers, considering queue backlog and delayed processing under the first-in-first-out (FIFO) queue discipline, can be modeled as the following set of linear constraints:

$$q_{t,u}^{(0)} = \lambda_{t,u} + \sum_{v:(v,u) \in \mathcal{E}} y_{t,v \rightarrow u}^{(0)}, \quad \forall t \in \mathcal{T}, u \in \mathcal{U} \quad (12a)$$

$$q_{t,u}^{(d)} = r_{t-1,u}^{(d-1)} + \sum_{v:(v,u) \in \mathcal{E}} y_{t,v \rightarrow u}^{(d)}, \quad \forall d \in \mathcal{D} \setminus \{0\}, t \in \mathcal{T} \setminus \{0\}, u \in \mathcal{U} \quad (12b)$$

$$q_{0,u}^{(d)} = 0, \quad \forall d \in \mathcal{D} \setminus \{0\}, u \in \mathcal{U} \quad (12c)$$

$$r_{t,u}^{(d)} = q_{t,u}^{(d)} - p_{t,u}^{(d)} - \sum_{v:(u,v) \in \mathcal{E}} y_{t,u \rightarrow v}^{(d)}, \quad \forall d \in \mathcal{D}, t \in \mathcal{T}, u \in \mathcal{U} \quad (12d)$$

$$r_{t,u}^{(d)} \geq 0, \quad \forall t \in \mathcal{T}, d \in \mathcal{D}, u \in \mathcal{U} \quad (12e)$$

$$r_{t,u}^{(D)} = 0, \quad \forall t \in \mathcal{T}, u \in \mathcal{U} \quad (12f)$$

The following are the w^{dc} decision variables. $q_{t,u}^{(d)}$ is the volume of tasks awaiting processing in the d -th buffering tier, $r_{t,u}^{(d)}$ the residual volume after processing, $p_{t,u}^{(d)}$ the task processing volume, and $y_{t,u \rightarrow v}^{(d)}$ the task migration volume between nodes, all in units of (tasks/timeslot). Parameters: s is the data volume coefficient per unit task, in units of (Gbps·timeslot/tasks); $\bar{B}_{t,uv}$ is the link bandwidth upper bound, in units of (Gbps); $\mu_{\max,u}$ is the maximum processing capacity of a single node, in units of (tasks/timeslot). $\lambda_{t,u}$ is the task arrival volume, in units of (tasks/timeslot), provided by the forecasting model $m_{\theta}^{dc}(\cdot)$.

Fig. 4 illustrates the temporal and spatial task transfer process in data centers under the FIFO queue discipline. For ease of description, spatial task transfer is only generated in the first period, with tasks originating from data center v . Temporal transfer is realized via the state transition equation (12b). The meanings of the constraints are as follows. Equation (12a) is the inflow balance of tier-0 buffering, consisting of local task arrivals and inbound migrated tasks. Equation (12b) is the task backlog state transition: after one time step of queuing, tier- $(d-1)$ buffering becomes tier- d buffering, with inbound migration superimposed; Online DFLOD must pass $r_{t,u}^{(d-1)}$ to the next window. Equation (12c) specifies the initial condition for non-zero buffering tiers, indicating that no non-zero-tier backlog exists initially. Equation (12d) is the post-processing residual balance, which can be substituted into (12b). Equation (12e) is the non-negativity constraint. Equation (12f) is the service level agreement (SLA) constraint: the maximum queuing delay of a task must not exceed D time steps, meaning that tasks reaching tier D must be cleared within the current period and must not carry over to the next period.

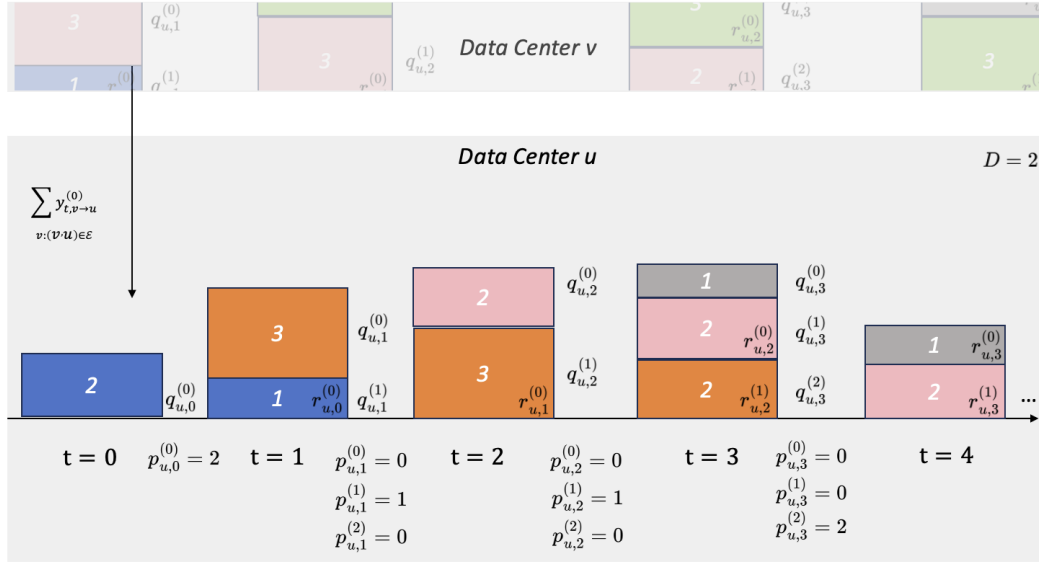


Figure 4. Schematic diagram of the spatiotemporal task transfer model in data centers under FIFO queue discipline

$$\begin{aligned}
 q_{T-1,u}^{(d)} &= q_{0,u}^{(d)}, & \forall d \in \mathcal{D}, u \in \mathcal{U} & \quad (13a) \\
 s \sum_{d \in \mathcal{D}} y_{t,u \rightarrow v}^{(d)} &\leq \bar{B}_{t,uv}, & \forall (u,v) \in \mathcal{E}, t \in \mathcal{T} & \quad (13b) \\
 p_{t,u}^{(d)} &\geq 0, & \forall d \in \mathcal{D}, t \in \mathcal{T}, u \in \mathcal{U} & \quad (13c) \\
 y_{t,u \rightarrow v}^{(d)} &\geq 0, & \forall d \in \mathcal{D}, t \in \mathcal{T}, (u,v) \in \mathcal{E} & \quad (13d) \\
 \lambda_{t,u} &\geq 0, & \forall t \in \mathcal{T}, u \in \mathcal{U} & \quad (13e) \\
 \sum_{d \in \mathcal{D}} p_{t,u}^{(d)} &\leq \mu_{\max,u}, & \forall t \in \mathcal{T}, u \in \mathcal{U} & \quad (13f)
 \end{aligned}$$

The meanings of these constraints are as follows. Equation (13a) is the cycle constraint, requiring that the task volumes in each buffering tier at the end of the period equal those at the beginning, used to decouple DA-DFLOD training rounds; slack variables can be introduced during training to ensure feasibility. Equation (13b) is the communication bandwidth constraint: the total data volume of outgoing tasks must not exceed the link bandwidth upper bound. Equations (13c)–(13e) are non-negativity constraints. Equation (13f) is the processing capacity constraint: the total processing volume of a single node in a single period must not exceed its maximum processing capacity. All constraints in this subsection together constitute the data center feasible region $\mathcal{W}^{dc}$. Through the FIFO discipline and state transition equations, the spatiotemporal transfer characteristics of data center tasks are formulated as a set of linear constraints, which helps avoid excessive computational burden across multiple training rounds. More importantly, this feasible region preserves the end-to-end mapping relationship among forecasted task arrivals, task migration decisions, and energy costs, enabling interconnected data center task scheduling to participate as the DFL lower-level optimization layer in day-ahead training and intraday delayed training, rather than serving as an isolated load preprocessing step.

3.3.3 Renewable Generation Constraints

Let $P_{t,u}^R$ be the available renewable generation of node u in period t , provided by the renewable generation forecasting model $m_{\theta}^R(\cdot)$. The renewable-side decision variables are denoted as $w_{t,u}^R = \{w_{t,u}^{R,use}, w_{t,u}^{R,cur}\}$, where $w_{t,u}^{R,use}$ is the actually consumed renewable power and $w_{t,u}^{R,cur}$ is the curtailed renewable power. The renewable generation feasible region $\mathcal{W}^R$ is defined as:

$$w_{t,u}^{R,use} + w_{t,u}^{R,cur} = P_{t,u}^R, \quad \forall t \in \mathcal{T}, u \in \mathcal{U} \quad (14a)$$

$$w_{t,u}^{R,use} \geq 0, \quad w_{t,u}^{R,cur} \geq 0, \quad \forall t \in \mathcal{T}, u \in \mathcal{U} \quad (14b)$$

Equation (14a) states that the available renewable generation is allocated between actual consumption and curtailment; Equation (14b) enforces non-negativity. If node u is not equipped with renewable generation, set $P_{t,u}^R = 0$, and the above constraints yield $w_{t,u}^{R,use} = w_{t,u}^{R,cur} = 0$. In the system power balance and objective function, $w_{t,u}^{R,use}$ serves as a local power source offsetting the data center electricity demand, and $w_{t,u}^{R,cur}$ measures the un-consumed renewable energy; when the objective function includes renewable consumption benefits or curtailment penalties, this constraint ensures that the dispatch result does not exceed the available renewable generation.

3.4 Forecasting Model

The forecasting model $m_\theta(\cdot)$ is used to map operational observation information to the uncertain parameters required by the lower-level scheduling. Rather than designing separate complex temporal models for individual forecasting targets, this paper adopts a unified parameterized forecasting structure that simultaneously outputs computing task arrivals, available renewable power, and electricity price parameters:

$$\hat{c}_{t,u}^a = m_{\theta^a}^a(x_{t,u}^a) = \mathcal{T}_a \left((x_{t,u}^a)^\top \theta^a + \theta_b^a \right), \quad a \in \{dc, R, \pi\}. \quad (15)$$

Here, a denotes the forecasting module category, corresponding to computing task arrivals, available renewable power, and electricity prices, respectively; $x_{t,u}^a$ is the input vector for period t and node u ; θ^a and θ_b^a are the corresponding parameter block and bias term, with the overall parameter vector $\theta = [(\theta^{dc})^\top, (\theta^R)^\top, (\theta^\pi)^\top]^\top$. $\mathcal{T}_a(\cdot)$ denotes scale restoration and physical boundary handling: for task arrivals and available renewable power, the output must be clipped to be non-negative and satisfy the upper bounds given by equipment or data; for electricity prices, $\mathcal{T}_\pi(\cdot)$ applies a logarithmic-exponential scaling of the price time series, preserving negative prices and price spikes.

The three types of forecast parameters collectively form the exogenous inputs to the scheduling model within the rolling window. The computing task forecasting module takes input $x_{t,u}^{dc} \in \mathbf{R}^{14}$ and outputs the task arrival volume $\hat{\lambda}_{t,u}$ for node u in period t ; the renewable forecasting module takes input $x_{t,u}^R \in \mathbf{R}^{11}$ and outputs the upper bound of renewable power that can be dispatched and consumed, $\hat{P}_{t,u}^R$; the electricity price forecasting module takes input $x_{t,u}^\pi \in \mathbf{R}^{30}$ and outputs the market electricity purchase and sale price parameters $\hat{\pi}_{t,u}^{buy}, \hat{\pi}_{t,u}^{sell}$. Within a rolling window of length H , the model generates

$$\hat{c}_{t:t+H-1} = \left\{ \hat{\lambda}_{\tau,u}, \hat{P}_{\tau,u}^R, \hat{\pi}_{\tau,u}^{buy}, \hat{\pi}_{\tau,u}^{sell} \right\}_{\tau=t}^{t+H-1}, \quad (16)$$

and feeds them into the power balance, energy storage, task migration, and renewable consumption constraints established in Sections 3.1–3.3, yielding the forecast-induced scheduling decisions. The detailed construction and standardization of these input features are provided in the Supplementary Material.

This paper simplifies the spatial meaning of the NWP features in the renewable forecast inputs. The NWP array is arranged along the three dimensions of “time period, spatial grid point, channel,” denoted as $Z_{t,i,j,k}^R$, with dimensions $T \times 11 \times 11 \times 2$, where T is the number of 15 min periods: the complete array covers 2023-01-01 to 2025-12-31 with 105216 rows; the 11×11 entries are meteorological grid points at 0.1° resolution, and the last dimension of size 2 contains the radiation flux and temperature elements, respectively. (2,2) and (10,10) are the positions in the NWP grid of the stations used to construct the renewable output sequences of the two nodes. The research focus of this paper is the decision-focused online learning mechanism itself rather than PV forecast accuracy; a complete per-node spatial meteorological input (multi-grid-point aggregation or independent NWP features for each node) is left for future work.

On the other hand, this paper also considers a nonlinear predictor with a ResNet structure, whose architecture, parameter selection, and initialization follow exactly the residual neural network (ResNet) used for storage-arbitrage price forecasting in³⁴. The network is stacked from several fully connected (FC) layers and residual connected (RC) layers, and the network mapping is:

$$\begin{aligned} y_{t,u}^{a,1} &= \text{ReLU} \left(\theta^{a,1} x_{t,u}^a + \theta_b^{a,1} \right) \\ y_{t,u}^{a,2} &= \text{ReLU} \left(\theta^{a,2} y_{t,u}^{a,1} + \theta_b^{a,2} \right) \\ \hat{c}_{t:t+H-1} &= \text{Broadcast} \left(x_{t,u}^a[-1] \right) + y_{t,u}^{a,2} \end{aligned} \quad (17)$$

The nonlinear predictor is shown in Fig. 5. The FC-layer parameters are randomly initialized, and the RC-layer parameters are initialized by the least-squares estimate $((X^a)^\top X^a)^{-1} (X^a)^\top Y^a$, where X^a and Y^a are the input sample matrix and output

sample vector of module a , respectively, so as to stabilize the training start of the deep network; for the price module, the output is set to the logarithmic form of the electricity price to stabilize training under the high volatility of prices. Since the training framework of this paper is insensitive to the predictor structure, any predictor is admissible as long as $\partial \hat{c} / \partial \theta$ can be computed or estimated.

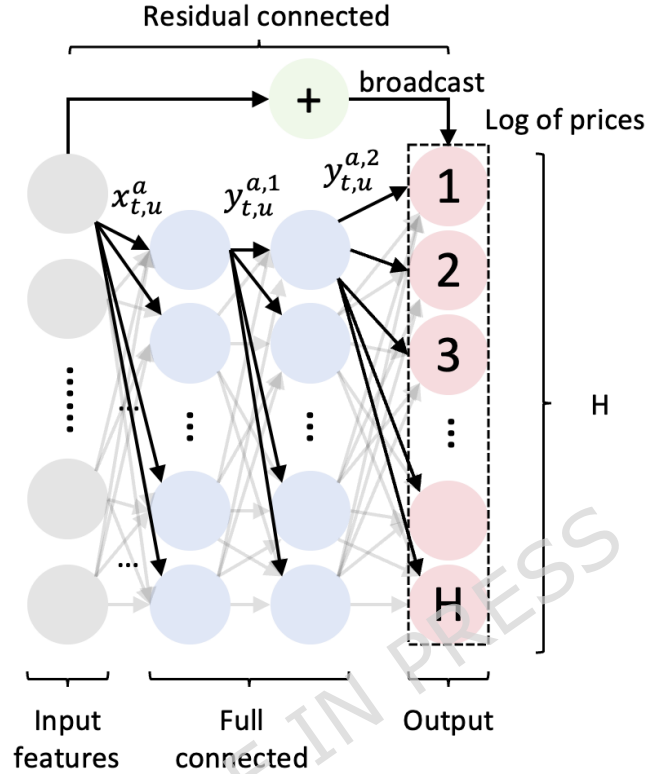


Figure 5. Nonlinear forecasting model with ResNet structure

The main experiment adopts the linear predictor to highlight the decision-focused online learning mechanism itself and obtain interpretable feature weights; the consideration of the nonlinear predictor in this paper shows that the proposed decision-focused online learning framework is general across predictors with different representation capacities, and systematic tuning and structural comparison of nonlinear predictors are left for future work.

4 Training of Decision-Focused Learning and Solution of Optimal Scheduling

The scheduling of interconnected data centers with distributed PV, energy storage, and task migration constraints simultaneously faces four types of difficulties: the multi-source uncertainties of electricity prices, PV generation, and task arrivals cannot be adequately evaluated solely by forecast error; cross-center task migration, storage state of charge, and queue backlogs cause forecast errors to affect subsequent scheduling through state transitions; information is gradually revealed during intraday operation, and day-ahead offline models cannot adapt to short-term disturbances in a timely manner; and the storage charging–discharging complementarity constraint introduces non-differentiability and numerical oscillations in the backward update of decision-focused learning. To address these issues, this section presents specific treatments at the training and solution levels. First, in the day-ahead stage, the power balance, energy storage, spatiotemporal task transfer, and renewable constraints from Section 3 are uniformly embedded in the lower-level scheduling, and the forecast parameters are trained using the upper-level posterior decision cost rather than MSE, so that the model learns “which forecast errors will actually change the operational cost.” Second, in the intraday stage, rolling execution and delayed training are adopted: at each time step, only the first action of the current window is executed, and parameters are updated after the true electricity prices, PV generation, and task arrival information is revealed, thereby making the learning process consistent with the actual operational closed loop of “decide first, observe later, then correct.” Finally, to address the complementarity constraint and the non-differentiability of the inner-level optimal solution mapping, this section employs convex relaxation or regularized feasible regions, a quadratic smoothing surrogate decision layer, constraint violation penalties, and SPSA zeroth-order gradient estimation, so that the training process

does not rely on analytic gradients, and feasibility checks are used to suppress the contamination of online parameters by infeasible solutions.

4.1 Day-Ahead Stage

4.1.1 Training of Day-Ahead Decision-Focused Learning

The role of DA-DFLOD is to learn a stable set of initial forecasting parameters over a long historical sample period, providing a prior for intraday rolling scheduling. The basic idea is to jointly input electricity prices, task arrivals, and renewable generation as forecast parameters into the lower-level scheduling, and back-propagate the training signal using the cost of forecast-induced scheduling under the true parameters. Thus, the model does not pursue the minimization of the statistical error of any single class of variables in isolation; instead, it preferentially corrects those forecast directions that would lead to high-cost deviations in electricity purchase/sale, storage charging/discharging, task migration, or renewable curtailment decisions. Let the i -th day-ahead training sample cover a complete scheduling window (whose number of periods is denoted by N_i), with true parameters $c_i = \{\pi_i^{\text{buy}}, \pi_i^{\text{sell}}, \lambda_i, P_i^R\}$ and forecast parameters $\hat{c}_i = m_\theta(X_i)$. Incorporating the power balance, energy storage, task migration, and renewable constraints established in Section 3, the day-ahead decision layer is written as

$$\begin{aligned} w_i^*(\hat{c}_i) \in \underset{w_i}{\operatorname{argmin}} \quad & J(w_i, \hat{c}_i) \\ \text{s.t.} \quad & w_i \in \mathcal{W}_i^P(\hat{c}_i) \cap \mathcal{W}_i^{s,\text{rel}} \cap \mathcal{W}_i^{dc}(\hat{\lambda}_i) \\ & \cap \mathcal{W}_i^R(\hat{P}_i^R). \end{aligned} \quad (18)$$

where $\mathcal{W}_i^P$, $\mathcal{W}_i^{s,\text{rel}}$, $\mathcal{W}_i^{dc}$, and $\mathcal{W}_i^R$ correspond to Equations (7b)–(7c), the storage convex relaxed feasible region, Equations (12a)–(13f), and Equations (14a)–(14b), respectively. The outer-level loss of day-ahead training uses the posterior cost of the forecast-induced scheduling under the true parameters as the objective:

$$\mathcal{L}_{\text{DA}}(\theta) = \frac{1}{|\mathcal{S}_{\text{tr}}|} \sum_{i \in \mathcal{S}_{\text{tr}}} [J(w_i^*(m_\theta(X_i)), c_i) - J(w_i^*(c_i), c_i)]. \quad (19)$$

In (19), $J(w_i^*(c_i), c_i)$ is the posterior optimal cost under full information, which is only used for normalization and regret evaluation and does not participate in actual operation. Since the full-information benchmark cost $J(w_i^*(c_i), c_i)$ can be regarded as a constant for each training sample, if one only compares the downward trend across different training rounds, $J(w_i^*(m_\theta(X_i)), c_i)$ can also be directly used as the training loss.

The training samples are partitioned chronologically into a training set $\mathcal{S}_{\text{tr}}$, a validation set $\mathcal{S}_{\text{val}}$, and a test set $\mathcal{S}_{\text{te}}$. The mean, standard deviation, and power baseline values of all input features are computed solely on $\mathcal{S}_{\text{tr}}$ and fixed for the validation set, test set, and intraday online stage, avoiding future information leakage. The training set is used to update θ ; the validation set is used to select the learning rate, smoothing coefficient, perturbation radius, early-stopping round, and regularization parameters; the test set is used for offline evaluation of the day-ahead stage, while the intraday online interval (June–December 2025) is used to simulate real optimal scheduling and report scheduling cost, forecast error, and constraint violations.

4.1.2 Regularization of the Energy Storage Constraint

For the current test case, when the storage subproblem satisfies the sufficient conditions for optimal convex relaxation in the literature, Equation (18) is solved using $\mathcal{W}_i^{s,\text{rel}}$. If the electricity sale mechanism causes these conditions to fail, the complementarity structure is retained and a regularized feasible region is adopted. For any $t \in \mathcal{T}$, $u \in \mathcal{U}$, the piecewise smooth function

$$\Phi_{t,u}(w^c, w^d, \varepsilon) = \begin{cases} (w^c - \varepsilon)(w^d - \varepsilon), & w^c + w^d \geq 2\varepsilon, \\ -\frac{(w^c - \varepsilon)^2 + (w^d - \varepsilon)^2}{2}, & w^c + w^d < 2\varepsilon \end{cases} \quad (20)$$

replaces $w_{t,u}^c w_{t,u}^d = 0$, yielding

$$\mathcal{W}_i^{s,\text{reg}}(\varepsilon) = \left\{ w : \begin{aligned} & \Phi_{t,u}(w_{t,u}^c, w_{t,u}^d, \varepsilon) \leq 0, \\ & 0 \leq w_{t,u}^c \leq \bar{w}_u^c, \\ & 0 \leq w_{t,u}^d \leq \bar{w}_u^d, \\ & \forall t \in \mathcal{T}, u \in \mathcal{U} \end{aligned} \right\}, \quad (21)$$

while retaining the energy recursion and energy bound constraints. As $\varepsilon \rightarrow 0^+$, $\mathcal{W}^{s,\text{reg}}(\varepsilon)$ shrinks to the original complementarity feasible region; in actual computation, the simultaneous charging–discharging violation is checked via $\max_{t \in \mathcal{T}, u \in \mathcal{U}} w_{t,u}^c w_{t,u}^d$ and $\max_{t \in \mathcal{T}, u \in \mathcal{U}} \Phi_{t,u}$.

4.1.3 Regularization of the Loss Function

To improve the numerical stability of the outer-level parameter update, this paper constructs a surrogate decision layer with a quadratic smoothing term during the training phase:

$$\tilde{w}_i(\theta) = \underset{w_i \in \tilde{\mathcal{W}}_i}{\operatorname{argmin}} \left\{ J(w_i, m_\theta(X_i)) + \frac{\alpha_k}{2} \|w_i\|_2^2 \right\}, \quad (22)$$

where $\tilde{\mathcal{W}}_i$ denotes the scheduling feasible region after convex relaxation or regularization, and $\alpha_k > 0$ is the objective-side smoothing coefficient. The quadratic term is only used in the training surrogate problem and is not counted in the true operational cost defined by (7a). The surrogate function value is defined as

$$L_i(\theta) = \frac{J(\tilde{w}_i(\theta), c_i) - J(w_i^*(c_i), c_i)}{s_J} + \rho \left(v_i^E(\theta) + v_i^\Phi(\theta) + v_i^{dc}(\theta) \right), \quad (23)$$

where s_J is a cost normalization factor, ρ is the infeasibility penalty coefficient; v_i^E , v_i^Φ , and v_i^{dc} denote the numerical violations of the storage energy bound, complementarity regularization, and task queue constraints, respectively. This penalty term prevents the outer-level perturbation from misclassifying a slightly infeasible inner-level solution as a better parameter.

4.1.4 Zeroth-Order Gradient Estimation

Since general-purpose optimization solvers typically do not directly provide $\partial \tilde{w}_i / \partial \theta$, day-ahead training employs zeroth-order gradient estimation. When the parameter dimension is small, dimension-wise finite differences can be used; when θ comprises the three parameter blocks for computing tasks, renewable generation, and electricity prices, the number of inner-level solves required by dimension-wise finite differences grows linearly with the dimension. Therefore, this paper adopts SPSA (simultaneous perturbation stochastic approximation) with random simultaneous perturbations. Let p be the parameter dimension, $\Delta \in \{-1, 1\}^p$ be a Rademacher random direction, and c_k be the perturbation radius; then

$$\theta_k^+ = \Pi_\Theta(\theta_k + c_k \Delta), \quad \theta_k^- = \Pi_\Theta(\theta_k - c_k \Delta), \quad (24)$$

$$\hat{\nabla}_j L(\theta_k) = \frac{L(\theta_k^+) - L(\theta_k^-)}{\theta_{k,j}^+ - \theta_{k,j}^-}, \quad \left| \theta_{k,j}^+ - \theta_{k,j}^- \right| > 0. \quad (25)$$

Estimates from multiple random directions can be averaged. Subsequently, the projected gradient update is applied:

$$\theta_{k+1} = \Pi_\Theta \left(\theta_k - \eta_k \hat{\nabla} L(\theta_k) \right), \quad (26)$$

where Π_Θ denotes projection onto the parameter box or norm constraints, and η_k is the learning rate. After each training round, the average decision loss, average regret, MSE, and constraint violations are recorded on $\mathcal{S}_{\text{tr}}$ and $\mathcal{S}_{\text{val}}$, respectively; the training loss curve is used to observe convergence, and the validation loss curve is used to judge overfitting. Training stops when the change in regret between adjacent rounds of the validation decision loss remains below the threshold $\text{tol} = 10$ for five consecutive rounds, and the θ_{DA} with the lowest validation loss is saved as the initial parameter for the intraday online stage.

The day-ahead training procedure is given in Algorithm 1 in Section 2.1. In the algorithm, “execution scheduling” refers to the scheduling problem without the quadratic smoothing term, and “surrogate scheduling” refers to Equation (22).

If the inner-level scheduling adopts convex relaxation, it can be solved by linear programming or quadratic programming solvers; when the regularization shown in (20) is incorporated, the inner-level problem becomes an optimization problem with smooth nonlinear constraints and is solved using gradient-based nonlinear programming solvers. In the implementation, the solution from the previous training window or the previous perturbation point is used as a warm start, and the power balance residual, energy bound violation, task queue violation, $\Phi_{t,u}$ violation, and objective function value are jointly used to determine whether the solution is acceptable.

4.2 Intraday Stage

4.2.1 Rolling Decision of Intraday Decision-Focused Online Learning

Online DFLOD performs rolling execution and delayed training on the basis of the day-ahead parameters θ_{DA} , addressing the closed-loop problem of “gradual intraday information revelation, continuous state transition, and immediate decision execution.” Compared with the day-ahead stage, the intraday stage no longer solves full-year or full-day samples in one shot; instead, at each actual operating time step t , it only uses the observed information $\mathcal{I}_t$ to generate H -step-ahead forecasts and treats the physical state passed from the previous window as the initial condition. The state variables include the storage state of charge E_t^{phys} and the task backlog $r_{t,u}^{(d),\text{phys}}$ at each buffering tier of each data center. This treatment can explicitly bring the energy and queue consequences of already executed actions into the next optimization, so that short-term price spikes, PV shading, or task surges are not merely recorded as forecast errors but are transformed into states and parameter updates that can be corrected in subsequent windows.

In the rolling execution phase, given the current parameters θ_t , the intraday model generates

$$\hat{c}_{t:t+H-1} = m_{\theta_t}(\mathcal{I}_t), \quad (27)$$

and solves the window scheduling problem

$$\begin{aligned} w_{t:t+H-1}^{*,\text{re}}(\theta_t) &\in \underset{w}{\operatorname{argmin}} J(w, \hat{c}_{t:t+H-1}) \\ \text{s.t. } w &\in \mathcal{W}_{t:t+H-1}(E_t^{\text{phys}}, r_t^{\text{phys}}, \hat{c}_{t:t+H-1}). \end{aligned} \quad (28)$$

Only the first action $w_t = w_{t,0}^{*,\text{re}}(\theta_t)$ is actually executed, after which the states E_{t+1}^{phys} and r_{t+1}^{phys} are updated based on the actual execution results. The remaining actions within the window serve only as plans under the current information and are not directly executed. This allows continuous strategy correction as renewable generation, electricity prices, and task arrivals are gradually revealed.

4.2.2 Delayed Training of Intraday Decision-Focused Online Learning

In the delayed training phase, upon entering time t , the true parameters for period $t-1$ and earlier have been revealed. Let the start of the lagged training window be $s = t - H$, with the window covering $s : s + H - 1$. Intraday training uses the already revealed true parameters $c_{s:s+H-1}$ of this window to evaluate the true cost of the forecast-induced scheduling, constructing

$$\begin{aligned} L_s^{\text{ID}}(\theta) &= \frac{1}{s_J} \left[J(\tilde{w}_s(\theta), c_{s:s+H-1}) \right. \\ &\quad \left. - J(w_s^*(c_{s:s+H-1}), c_{s:s+H-1}) \right] \\ &\quad + \rho v_s(\theta), \end{aligned} \quad (29)$$

where $\tilde{w}_s(\theta)$ is obtained from the same surrogate scheduling as in (22), and $v_s(\theta)$ aggregates the power balance, energy storage, complementarity regularization, and queue constraint violations. Since the training window always lags behind the execution window, Equation (29) does not use future true information that has not yet occurred.

4.2.3 Parameter Update of Intraday Decision-Focused Online Learning

Gradient propagation in intraday rolling training adopts a “state truncation, parameter delay” approach. State truncation means that E_s^{phys} and r_s^{phys} enter the training window as the physical initial values of the already executed system, and backpropagation to earlier-period actions is not continued; parameter delay means that the gradient obtained from window $s : s + H - 1$ is only used to update parameters for the next or subsequent time steps, without altering the already executed w_t . Therefore, the outer-level update can be written as

$$\begin{aligned} \vartheta_s &= \Pi_{\Theta} \left(\theta_t - \eta_t \widehat{\nabla} L_s^{\text{ID}}(\theta_t) \right), \\ u_s &= \vartheta_s + \delta_s(\theta_t - \vartheta_s), \quad \delta_s \sim \text{Unif}[0, 1], \\ \theta_{t+1} &= \Pi_{\Theta} \left(\theta_t - \eta_t \widehat{\nabla} L_s^{\text{ID}}(u_s) \right). \end{aligned} \quad (30)$$

Equation (30) is consistent with the delayed training framework presented in Section 2. ϑ_s denotes the temporary approximate oracle parameter induced by the most recently revealed window, and u_s is used to randomly interpolate between the current parameter and the temporary parameter, reducing the update oscillations caused by online objective drift. $\widehat{\nabla} L_s^{\text{ID}}$ is likewise

estimated via SPSA; the intraday stage requires only a constant number of surrogate scheduling solves per time step, making it suitable for rolling operation.

The input features of the intraday stage follow the computing task, renewable, and electricity price features defined in Section 3, but the available information set is updated over time. For electricity prices and regional demand, rolling means, rolling standard deviations, and lag terms only use observations up to and including t ; for renewable generation, the intraday stage can receive the latest NWP or short-term forecasts and reconstruct $X_{t:t+H-1}^R$; for computing tasks, the task arrival lag terms and queue backlog states are refreshed synchronously. All feature standardization still uses the training-set statistics from the day-ahead stage, avoiding parameter-scale mutations due to short-window mean drift in the online stage.

The intraday solution procedure is given in Algorithm 2 in Section 2.2. Unlike the day-ahead stage, the intraday stage focuses more on online stability than on single-round optimality. To suppress excessive parameter perturbations caused by price spikes or anomalous renewable forecasts, η_t decreases over time, the SPSA perturbation radius scales with $\|\theta_t\|_2$, and a projection constraint is imposed on θ_t . If the inner-level scheduling of a rolling window fails the feasibility check, the parameters from the previous time step are retained and the solution is reattempted with a more conservative warm start or a smaller perturbation radius, preventing infeasible gradients from contaminating the online model.

To cope with the disturbance of extreme events, the intraday stage further introduces anomaly identification and a safety mode. The identification rules are: the electricity price exceeds 15000 AUD/MWh (the order of magnitude of the Australian electricity spot market price cap) for no fewer than four consecutive periods; the available PV power drops by no less than 0.2 MW (20% of the rated capacity) per period for no fewer than four consecutive periods; or the task arrival volume $\lambda_{t,u}$ exceeds 80% of the node processing capacity μ_u for no fewer than two consecutive periods (the thresholds of the two nodes are 3.44 and 3.20 tasks/period, respectively). When any signal is triggered, the system suspends the parameter updates of delayed training and rolls back to the most recent valid parameters before the trigger; meanwhile, full-window optimization is stopped and a greedy strategy is executed instead: PV energy is consumed locally first, the storage charging/discharging thresholds are set according to the 25% and 75% quantiles of the training-period prices in each region, and tasks are processed locally in the queue within the SLA deadline. After the signal disappears and a re-check over four consecutive periods passes, online updates are resumed with a reduced learning rate. When physical-layer events such as grid faults occur, the optimization algorithm stops and the operator-defined safety strategy takes over (e.g., switching to storage and diesel generator supply through the STS); the emergency strategy after the switch is outside the scope of this paper.

Together, the day-ahead and intraday stages form the optimal scheduling training and solution framework of this paper. The day-ahead stage uses long-period historical data to learn seasonal and weekly periodic patterns and selects stable initial values through the validation set; the intraday stage uses the latest operational measurement data to correct short-term deviations, and ensures through delayed training that parameter updates only depend on revealed information. Finally, in Section 5, training loss curves, validation loss curves, economic performance on the day-ahead test set and the online interval, forecast MSE, and storage and queue constraint violations are reported simultaneously to validate the effectiveness of decision-focused learning relative to the conventional “predict-then-decide” pipeline.

5 Simulation Case Studies

5.1 Experimental Setup

The case study adopts a 15 min time resolution and selects two electricity price regions, NSW and VIC, to represent two data center nodes, with the latter exhibiting larger price volatility. It is assumed that the purchase and sale prices of electricity are equal. Electricity prices are taken from the Australian spot market and PV from a Chinese NWP station, so the seasonality is analyzed from the Northern Hemisphere perspective. Electricity prices and regional demand are sourced from public AEMO data, computing task arrivals are obtained from a locally preprocessed telecommunications-type task sequence, and available renewable power is constructed from actual PV generation data. The dataset is partitioned chronologically into a training set, a validation set, and a test set; the intraday online training and testing interval spans from June 1, 2025 to December 31, 2025. The electricity price distributions of the two regions are shown in Fig. 6; the VIC region exhibits higher price dispersion and spike risk, and can be used to test the adaptability of the online learning method under price volatility scenarios.

Two 2 MW data center nodes are set up in the two electricity price regions, respectively. The task arrival volume is evenly distributed according to node capacity, and the renewable installed capacity is 1 MW. The integrated storage–backup electrochemical battery has capacity bounds of 0 and 1 MWh, an initial energy of 0.5 MWh, charging and discharging power upper bounds of 1 MW, and charging/discharging efficiencies of 0.93. The task queue adopts the FIFO discipline, with a maximum buffering tier of 2, a maximum delay SLA of 30 min, single-period processing capacities of 4.3 and 4.0 for the two nodes, and an inter-node migration bandwidth upper bound of 2.0.

The day-ahead stage executes DA-DFLOD and day-ahead predict-then-decide (DA-PTD) scheduling computations without performing actual dispatch, using the results only as input to the intraday operation. DA-PTD uses MSE as the objective to obtain initial values, while DA-DFLOD uses SPSA to estimate the decision loss direction and adopts a regularized continuous

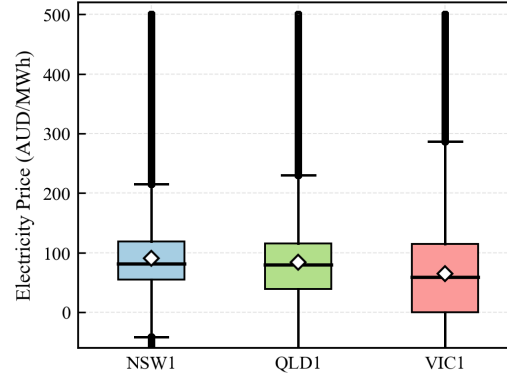


Figure 6. Electricity price characteristics of the NSW and VIC regions

Table 2. Model hyperparameters and equipment parameter settings

Category	Parameter	Value or Description
Learning model	Time resolution Δt	15 min
Learning model	Forecast window H	12 periods
Learning model	Day-ahead max training rounds K	400 rounds
Learning model	DFL learning rate η_k, η_t	Initial 0.018, decreasing with training rounds
Learning model	PTD learning rate	Initial 0.012, used for MSE baseline parameter updates
Learning model	SPSA perturbation radius c_k, c_t	0.04
Learning model	Surrogate smoothing coefficient α_k, α_t	1.0×10^{-5}
Learning model	Parameter projection range Θ	Each dimension constrained within $[-4, 4]$
Learning model	Regularization parameters ε, ρ	For complementarity regularization and infeasibility penalty, selected on validation set
Data/Features	Electricity price and load data	AEMO NSW, VIC regions, raw samples spanning 2023-01–2025-12, sampling interval 15 min
Data/Features	Dataset partition	Training set 2023-01-01–2024-03-31, validation set 2024-04-01–2024-10-31, test set 2024-11-01–2025-05-31, intraday online interval 2025-06-01–2025-12-31
Data/Features	NWP data	Center coordinates 108.1°E, 27.5°N, array dimensions $105216 \times 11 \times 11 \times 2$ (2023-01-01–2025-12-31, 15 min), with (2, 2) and (10, 10) as the forecast coordinates
Data/Features	NWP channel range	Two channel value ranges at the grid points of the two stations approximately $[0.1343, 0.8672]$ and $[0.0849, 0.3341]$
Data/Features	Input feature dimensions	Computing tasks 14-dim, available renewable power 11-dim, electricity prices 30-dim
Equipment/System	Data center nodes	NSW1, VIC1 two nodes, 2MW per node
Equipment/System	Renewable capacity	1 MW PV available power upper bound per node
Equipment/System	Storage capacity range	0–1 MWh, initial energy 0.5 MWh
Equipment/System	Storage power and efficiency	Charging and discharging upper bounds both 1 MW, $\eta_c = \eta_d = 0.93$
Equipment/System	Task queue	FIFO discipline, maximum buffering tier $D = 2$
Equipment/System	Single-period processing capacity	Two nodes 4.3 and 4.0 tasks/timeslot, respectively
Equipment/System	Inter-node migration bandwidth	2.0 Gbps

surrogate problem for parameter updates. Fig. 7 shows the variation of the one-day-averaged regret cost during the day-ahead training process (NSW region).

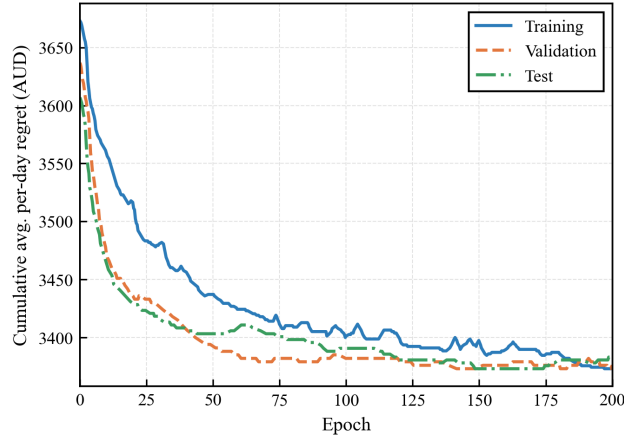


Figure 7. DA-DFLOD surrogate cost regret convergence curve

DA-DFLOD is considered converged when the change in regret between adjacent training rounds remains below the threshold $tol = 10$ for five consecutive rounds. Through multiple multi-region training runs, we found that training converges in approximately 175–200 rounds; with 15 months of training samples, the total runtime is about 3 hours. The training process was performed on a MacBook Pro with an M4 chip and 24 GB memory, with the maximum number of threads set to 7.

The intraday stage adopts the “rolling execution–delayed training” procedure. In each period, the scheduler forecasts the future $H = 12$ periods based on the current parameters and solves the complete window, but only executes the first action. After the true electricity price, task arrivals, and renewable generation of the next period are revealed, the system settles the cost using the true first-step parameters and updates the storage energy and task queue states. The Oracle benchmark solves using the true future window; intraday online predict-then-decide (Online PTD) solves using the forecasting parameters obtained from MSE training; Online DFLOD is initialized with the DA-DFLOD parameters and performs delayed updates through the regularized surrogate problem based on the revealed windows. Online DFLOD’s training stage adopts the regularized continuous surrogate; the actual execution scheduling of Oracle, Online PTD, and Online DFLOD all uses the Gurobi 13.0.0 MIP solver and enforces non-simultaneous storage charging and discharging through binary variable constraints.

5.2 Economic Comparison with Baseline Methods

We use the suffix RN to denote online decision-making with nonlinear forecasting methods. Table 3 presents the economic comparison for the June 2025 online test. Benefit is defined as the cost saving relative to Online PTD, i.e., $Benefit = J_{PTD} - J_{method}$; Regret is defined as the cost difference relative to Oracle, i.e., $Regret = J_{method} - J_{Oracle}$. Thus, a larger Benefit indicates greater improvement over Online PTD, and a smaller Regret indicates closer proximity to the full-information dispatch.

Table 3. Economic comparison of Online DFLOD for June 2025

Method	Total Cost	Benefit	Regret	MSE	Storage Constraint Violation
Oracle	213991.03	206639.20	0.00	0.00	0.00e+00
Online DFLOD	287272.96	133357.27	73281.93	417382.44	0.00e+00
Online DFLOD-RN	263592.91	157037.32	49601.88	405481.14	0.00e+00
Online PTD	420630.23	0.00	206639.20	97428.82	0.00e+00
Online PTD-RN	407820.59	12809.64	193829.56	87448.19	0.00e+00
MPC	401522.90	19107.33	187531.87	85877.39	0.00e+00

As can be seen from Table 3, Oracle, serving as the full-information benchmark for intraday rolling operation, achieves the lowest operating cost; Online DFLOD incurs a regret of 73281.93 relative to Oracle. Online PTD achieves a lower one-step-ahead MSE than Online DFLOD, but a higher total operating cost. This result indicates that, in a complete model incorporating task queues, renewable consumption, and storage complementarity constraints, merely reducing forecast error does not guarantee improved economic performance.

After introducing the ResNet nonlinear predictor, the total cost of Online DFLOD-RN is 263592.91, approximately 8.2% lower than that of Online DFLOD; the total cost of Online PTD-RN is 407820.59, approximately 3.0% lower than that of

Online PTD. The MSE of Online DFLOD-RN is 405481.14, only slightly lower than the 417382.44 of Online DFLOD; its forecast accuracy improvement is not significant, but the scheduling cost is about 8.2% lower than that of Online DFLOD, as analyzed in detail in the subsection “Influence of SPSA Randomness on the Results.” The MSE of Online PTD-RN decreases from the 97428.82 of Online PTD to 87448.19, with improved forecast accuracy and a simultaneous decrease in total cost. This shows that for Online PTD trained with an MSE objective, the nonlinear predictor and the logarithmic transformation of prices can capture part of the nonlinear fluctuation patterns in the price series, thereby improving price tracking ability to a certain extent and reducing operating cost.

Fig. 8 shows the Online DFLOD scheduling results for a typical day. Around 14:00, Online DFLOD provides the decision-maker with a low-then-high electricity price reference, whereas Online PTD provides a high-then-low reference. The former guides the solver to reserve storage in preparation for subsequent discharging; this forecasting behavior can be understood as delayed training adjusting the forecasting model’s judgment of the relative relationship in the price time series according to the observed scheduling cost feedback. In addition, we also find that in certain situations, Online DFLOD tends to raise the overall electricity price forecast, making the storage more cautious in its charging behavior and thereby avoiding substantial losses from misjudging prices. Online DFLOD-RN produces more accurate price forecasts, but since the relative magnitude of prices does not change much, the storage actions it induces do not have a significant cost advantage over those of Online DFLOD. On the other hand, the forecasts of Online PTD-RN are slightly improved and its cost is correspondingly lower than that of Online PTD, yet it still cannot match Online DFLOD’s cost optimization. These phenomena indicate that Online DFLOD, trained with the decision loss, adjusts the output of the forecasting model in a direction favorable to reducing scheduling cost, rather than merely pursuing pointwise forecast accuracy. The parameter weight changes in Section 5.4 confirm this observation at the feature level: DFL strengthens the electricity-price inertia and daily-periodic peak–valley signals while weakening the influence of short-term load noise, making the forecasting model output more conducive to storage charging/discharging timing decisions.

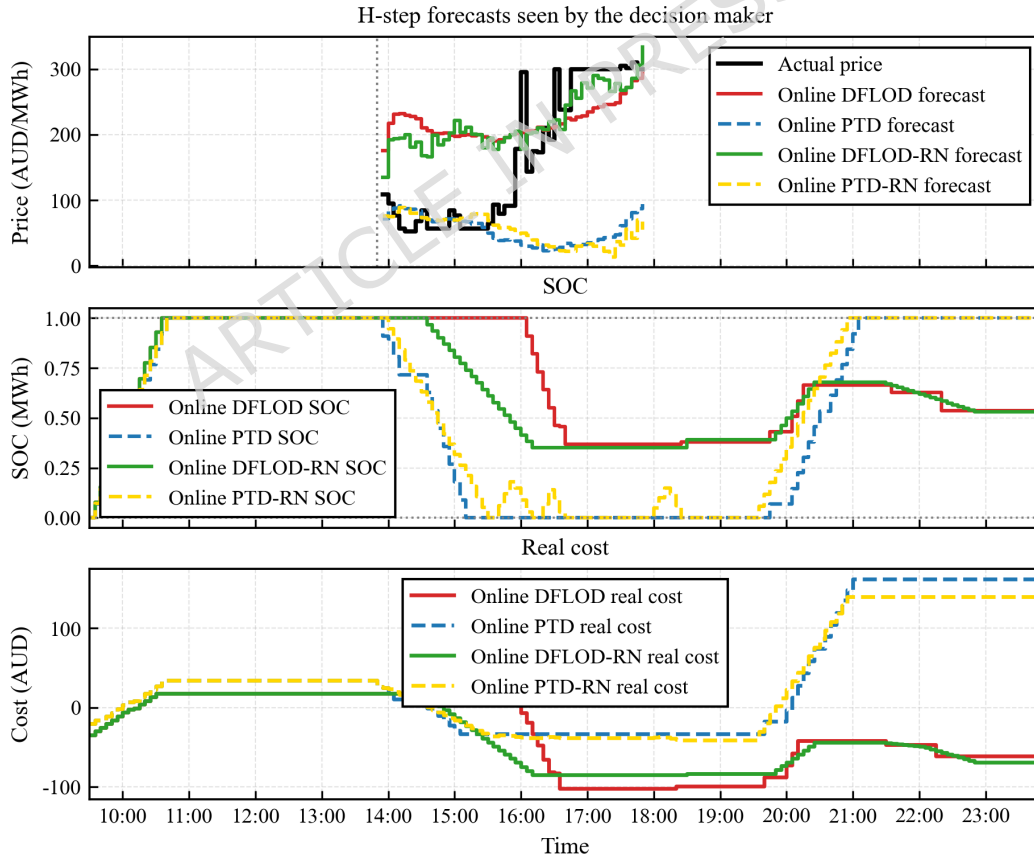


Figure 8. *H*-step look-ahead as perceived by the Online DFLOD decision-maker

Fig. 9 shows the solution results of intraday rolling and delayed training on June 29 of the online interval. In terms of electricity price forecasting, Online PTD always attempts to follow the true price with minimal MSE, but around noon (green

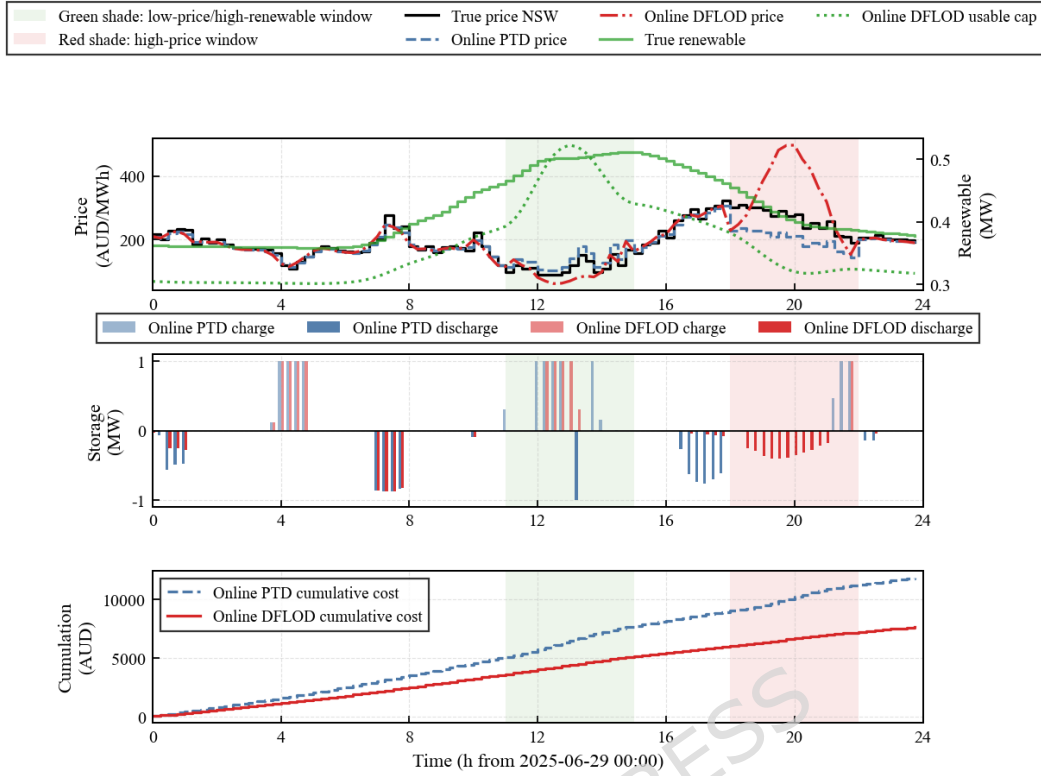


Figure 9. Online DFLOD scheduling results for data center energy storage

shaded interval) it forms a small peak–valley difference that causes the storage to discharge prematurely, thereby forfeiting the opportunity to discharge during the truly high-price interval. In contrast, Online DFLOD constructs a lower price in the green interval, suppressing the possibility of storage discharge and completing a storage reservation, thereby discharging during the red high-price interval and obtaining higher revenue.

Fig. 10 shows the task processing heatmap of data center node 1 during the online rolling process. At the beginning and end of the day, the residual tasks r are both zero; the task queue q is temporally shifted as much as possible to low-price periods, and task processing is concentrated between 10:00 and 16:00, when PV output is high. Tasks propagate between adjacent periods along with the queue state. Fig. 9 presents the electricity purchase, storage charging/discharging, and energy state of Online DFLOD on the test day, with no simultaneous storage charging–discharging violations observed. Storage energy bound violations, queue SLA violations, and power balance residuals are all zero.

Fig. 11 compares the data center energy costs of the five methods across different months. When solving across months, the rolling window is truncated within the current month to ensure comparability of costs across different months. As shown, Online DFLOD achieves lower energy costs than Online PTB in all months, validating the economic advantage of decision-focused learning in intraday rolling execution. In months with price spikes (e.g., June), Online PTB costs reach 420.63 k\$, which Online DFLOD reduces to 287.27 k\$, a reduction of 31.7%; in months with stable prices (e.g., October), the Online PTB cost is 120.22 k\$ and Online DFLOD is 81.25 k\$, with the absolute difference narrowing to 38.97 k\$. Overall, Online DFLOD reduces monthly costs by 16.5%–32.4% compared with Online PTB, and the cost curve of Online DFLOD consistently stays closer to the Oracle lower bound, indicating that the delayed training update mechanism effectively narrows the gap to the globally optimal solution.

Fig. 12 illustrates a price spike event in the NSW region in June. The first subplot shows the forecast results that the decision-maker can see at 15:00: with the linear predictors, neither Online DFLOD nor Online PTB achieves high accuracy in forecasting the price spike, while Online DFLOD-RN, with the ResNet nonlinear predictor, improves the price forecast accuracy slightly but not significantly. However, Online DFLOD's forecast exhibits a low-then-high trend; this relative high–low relationship of forecast values guides the storage to refrain from discharging at 15:00. This decision enables the storage to wait until the true price spike occurs before discharging, thereby allowing Online DFLOD to significantly outperform Online PTB in months with high electricity prices.

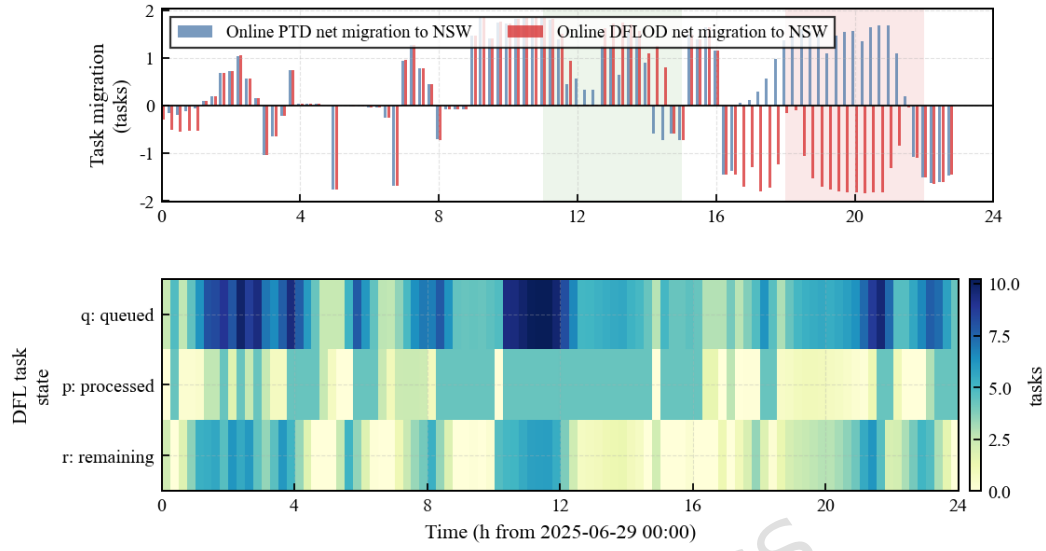


Figure 10. Online DFLOD scheduling results for data center task migration and task backlog

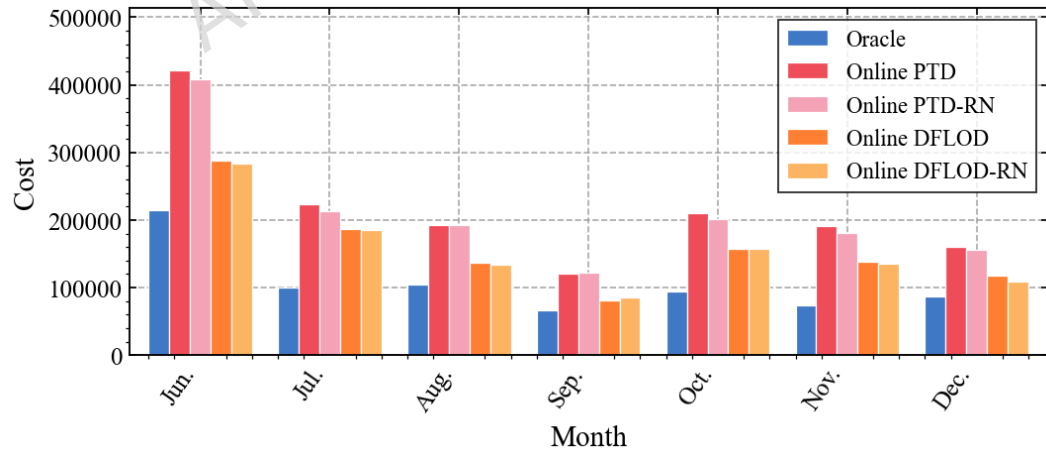


Figure 11. Monthly cost comparison of different methods

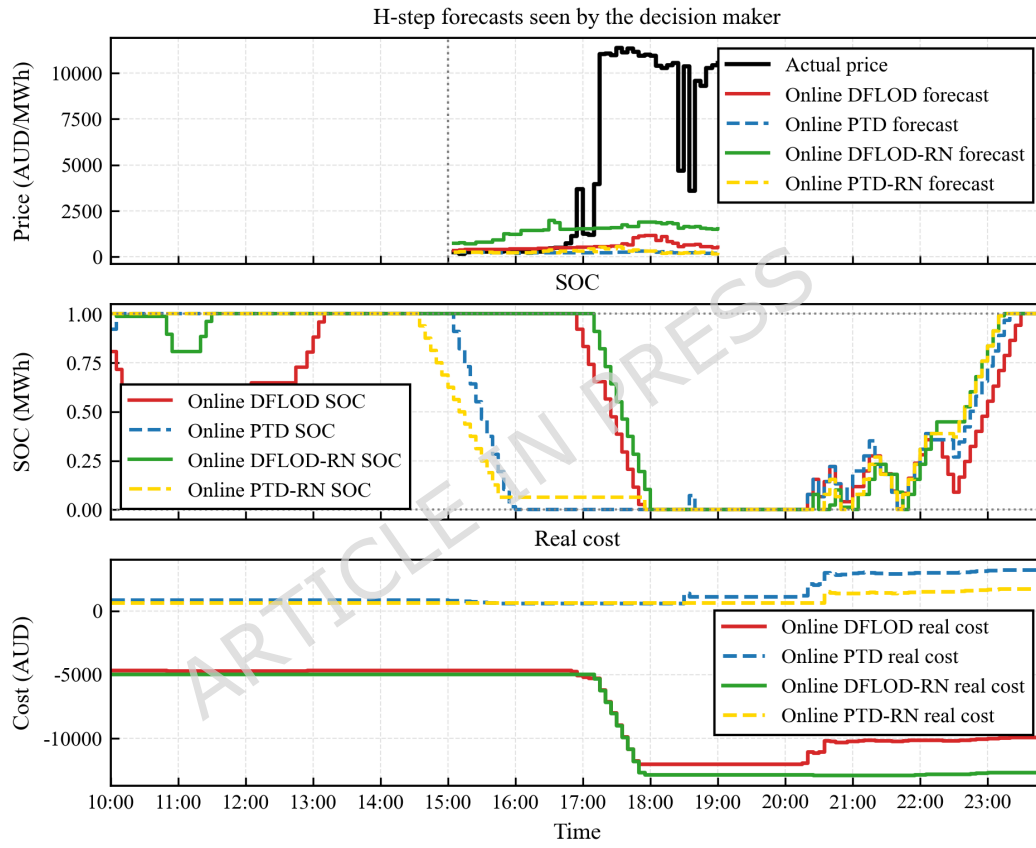


Figure 12. Performance comparison of Online DFLOD and Online PTD under a price spike

5.3 Improvement in PV Consumption

5.3.1 PV Consumption Performance

Assume the scenario does not permit data center PV reverse power flow, i.e., set $P_{t,u}^{\text{sell}} = 0$, in order to assess how much distributed PV can be consumed locally rather than exported. The PV consumption rate is defined as the ratio of locally consumed energy to total PV generation:

$$\eta = \frac{\sum_{t,u} w_{t,u}^{R,\text{use}} \Delta t}{\sum_{t,u} P_{t,u}^R \Delta t}, \quad (31)$$

where $P_{t,u}^R$ is the actual PV output of node u in period t , and $w_{t,u}^{R,\text{use}}$ is the locally consumed portion (including storage charging and direct use by data center loads).

Fig. 13 shows the PV consumption rates and consumption volumes of the five methods from June to December 2025. Overall, Online PTD exhibits the weakest consumption performance, with a cumulative consumption of 632.61 MWh over seven months and a total consumption rate of only 64.1%; Online DFLOD achieves a cumulative consumption of 772.47 MWh, with a total consumption rate raised to 78.3%; Oracle achieves a cumulative consumption of 821.12 MWh, with a total consumption rate of 83.3%. Online DFLOD improves the cumulative consumption by 139.86 MWh over Online PTD, fully demonstrating the effectiveness of decision-focused learning in enhancing local PV consumption.

On a month-by-month basis, the difference in consumption is most pronounced in the summer months (June–August) when PV output is abundant. In June, the total PV output is 149.79 MWh; Online PTD consumes only 99.21 MWh (66.2%), while Online DFLOD consumes 126.58 MWh (84.5%), an improvement of 18.3 percentage points; Oracle consumes 146.21 MWh (97.6%). In July, Online DFLOD achieves the highest consumption rate over the test period of 91.7% (150.89 MWh), significantly higher than Online PTD's 73.6% (121.06 MWh). In August, Online DFLOD consumes 143.47 MWh (87.3%), the month with the largest improvement over Online PTD, with the consumption rate exceeding by 22.8 percentage points.

Entering autumn, PV output weakens and the consumption rates of all methods generally decline, but Online DFLOD's advantage persists. In September, Online DFLOD consumes 107.14 MWh (71.0%), while Online PTD consumes 100.11 MWh (66.3%), with the gap narrowing to 4.7 percentage points. In October, the Online DFLOD consumption rate is 68.1% (94.31 MWh) versus Online PTD's 58.7% (81.22 MWh), with the gap widening again to 9.5 percentage points. In the winter months, December has the lowest PV output (106.90 MWh); in November, Online PTD's consumption rate drops to the annual minimum of 49.8% (55.50 MWh), while Online DFLOD still maintains 61.0% (67.96 MWh), 11.2 percentage points higher. In December, Online DFLOD consumes 82.13 MWh (76.8%) and Online PTD consumes 69.44 MWh (65.0%).

It should be noted that the full-information economic dispatch Oracle refers to the optimal scheduling that minimizes the comprehensive operating cost given knowledge of all true electricity prices and PV generation, rather than maximizing PV consumption. Therefore, Oracle's consumption rate is not 100%: during periods when storage is full, load is satisfied, and reverse power flow is not permitted, Oracle will, from a cost-optimal perspective, choose to curtail PV generation whose marginal value is negative. For example, in November, Oracle's consumption rate is only 67.6%, indicating that even with full information, economic dispatch itself involves a trade-off between economic efficiency and consumption rate. Online DFLOD's consumption rate lies between Online PTD and Oracle in all months and is closer to the Oracle level, validating the significant effectiveness of decision-focused learning in improving the PV consumption rate.

5.3.2 Market Policy Sensitivity Analysis

This section performs a sensitivity analysis with respect to the grid export permission. Equation (7a) is general with respect to whether reverse power flow is allowed: when selling is allowed, the sale revenue is settled at the regional spot price; when selling is prohibited, $P_{t,u}^{\text{sell}} = 0$. Taking the NSW and VIC spot regions as an example, the average spot prices of the two regions in 2025 are approximately 80–105 AUD/MWh; in contrast, the electricity exported to the grid by distributed PV is usually settled at the retailer feed-in tariff (FiT). The benchmark FiT range of NSW for 2025–26 is 4.8–7.3 c/kWh (IPART), and the minimum FiT of VIC since July 2025 is only 0.04 c/kWh (ESC), both far below the average spot price. Therefore, the equal purchase–sale price setting ($P_{t,u}^{\text{sell}} > 0$ settled at the spot price) constitutes an optimistic upper bound of the export permission, while prohibiting reverse power flow ($P_{t,u}^{\text{sell}} = 0$) corresponds to a conservative lower bound with the FiT approximately zero; together they form an envelope of the actual policies.

Fig. 14 and Fig. 15 present two results of this sensitivity analysis. Fig. 14 compares the final weights (7-month averages) of representative features of the DFL price-forecasting block under the two settings of allowing and prohibiting electricity sales; Fig. 15 shows the trajectory of the June price-inertia weight π_t during intraday online updates. The results show that the export permission setting has little influence on the feature-weight structure learned by DFL: among 7 months $\times$ 7 representative features, the maximum absolute weight difference between the two settings is 0.0173 and the average is only 0.0040, and the weights of features such as price inertia, the price at the same time on the previous day, and daily rolling statistics almost

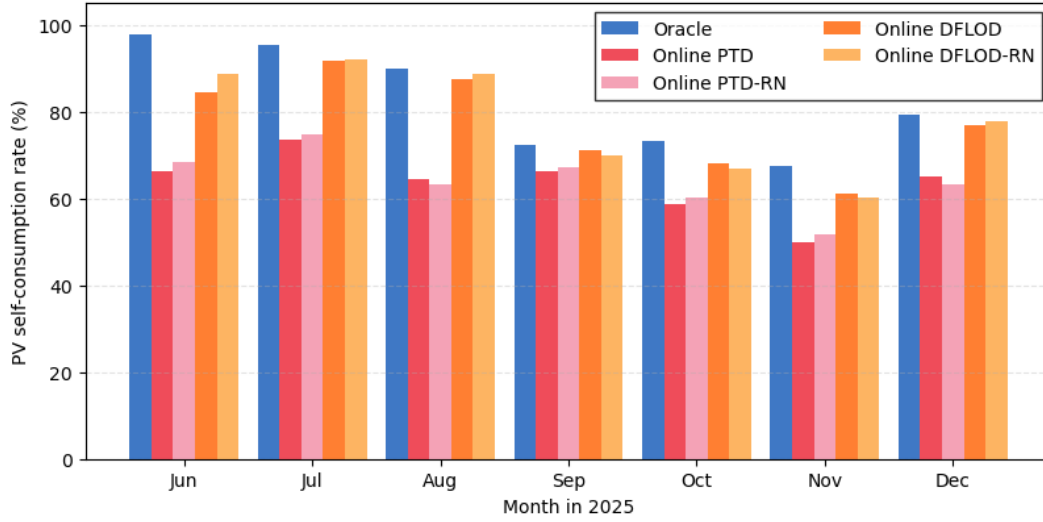


Figure 13. PV consumption rate comparison of different methods

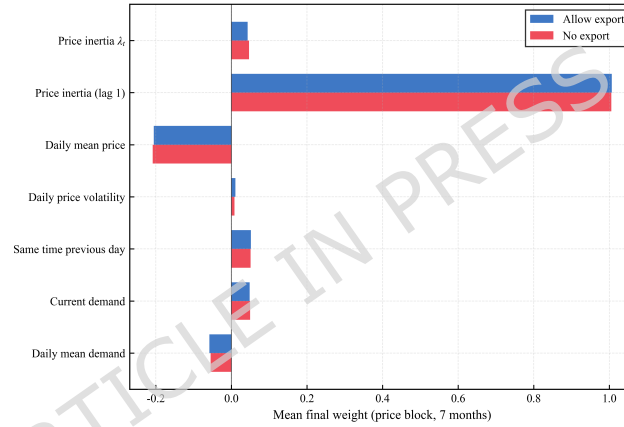


Figure 14. Comparison of DFL price-block feature weights under the sales-allowed and sales-prohibited settings (7-month average final values)

coincide under the two settings. This indicates that decision-focused learning captures feature structures with stable value for scheduling cost, and is robust to the export permission setting.

From the operational indicators, allowing electricity sales reduces the monthly total costs of Oracle and Online PTD by about 13% on average (PV surplus can be exported for revenue), while Online DFLOD decreases by only about 3.8%. Therefore, the scheduling behavior of Online DFLOD remains structurally robust to the export permission. In terms of the PV consumption rate, the difference between the two settings is only 1–3 percentage points; it should be noted that the consumption rate in the sales-allowed scenario is calculated according to the PV utilization volume (local consumption plus export), which differs from the accounting basis of Section 5.3.1 that only counts local consumption in the no-export scenario, so this difference mainly comes from the statistical basis rather than a substantive change in local consumption capability.

5.4 Analysis of Forecasting Parameters

Since the parameters of the linear regression model are directly the weights of the input features, one can directly observe how the model adjusts its reliance on different information sources during the online training process. From the parameter trajectory plot in Fig. 16, it can be seen that decision-focused learning does not cause all parameters to drift uniformly; rather, it mainly adjusts a small number of weights related to electricity price inertia, daily periodicity, and load state. Among them, π_t consistently maintains the largest positive weight, increasing from 0.2666 to 0.2794, indicating that the model continuously strengthens the continuation effect of the current electricity price on short-term forecast prices. Since in the storage scheduling objective the forecasted price carries a positive sign for charging power and a negative sign for discharging power, as this weight

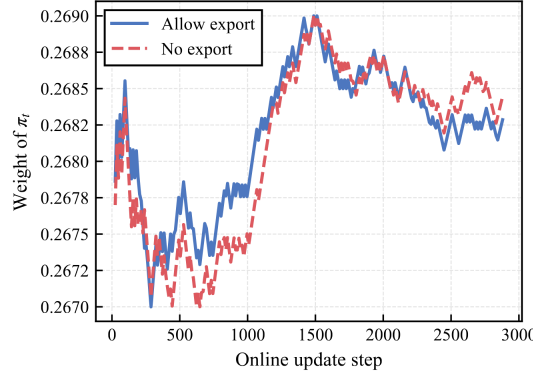


Figure 15. Online update trajectory of the price-inertia weight π_t in a typical month under two export permission settings

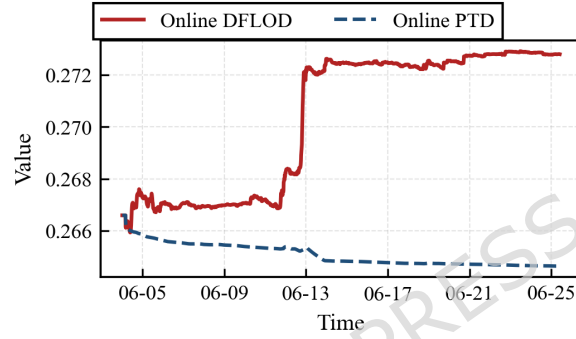


Figure 16. Variation of π_t weight with Online DFLOD training rounds

increases, high current prices are more readily propagated into the forecast window, thereby raising the charging cost and enhancing the discharging revenue, guiding the storage to reduce charging and favor discharging under high-price conditions.

More notably, the DFL weights of π_{t-288} (see Fig. 17) and d_{t-288} increase significantly, from 0.0079 to 0.0699 and from 0.0067 to 0.1581, respectively, whereas the final PTD values are only 0.0095 and 0.0181. This indicates that decision-focused learning places greater emphasis on the daily-periodic information of the same time three days earlier. For storage arbitrage, whether to charge or discharge in a given period depends primarily on the relative price spread and the peak–valley ordering within the forecast window, rather than on the MSE at individual time points. Daily-periodic electricity price and load information can help the model identify the recurring peak and valley segments each day; thus, the increase in their weights causes the forecasted prices to more readily form usable high–low price structures, thereby improving the charging/discharging timing.

In contrast, the DFL weight of d_t (see Fig. 18) decreases from 0.1275 to 0.0332, and that of d_{t-1} drops from near 0 to -0.0819 , whereas in PTD these two parameters change relatively little. This shows that DFL weakens the direct positive dependence on the current and immediately preceding load during training, preventing the model from excessively raising the forecasted price based on short-term load fluctuations. This adjustment helps reduce erroneous charging/discharging actions induced by local noise, causing the storage decisions to rely more on the daily-periodic peak–valley structure, which possesses stable decision value.

These parameter changes indicate that decision-focused learning does not simply pursue minimizing pointwise forecast error; rather, it actively strengthens the feature weights of those features that alter the optimal storage actions during training. The interpretability of the linear model makes this process directly observable: DFL enhances electricity price inertia and daily-periodic peak–valley signals while weakening the influence of short-term load noise, thereby transforming the forecasting model into a cost-signal generator more conducive to storage charging/discharging decisions.

5.5 Further Analysis

This section analyzes the proposed method from five perspectives, and the order of the subsections is the order of the discussion. The first is the computational cost analysis: the inner-level scheduling optimization of the proposed method is the main source of computational burden; we quantify the computational complexity and actual runtime of the day-ahead offline training

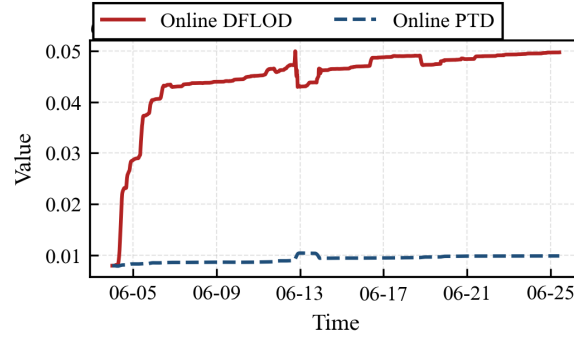


Figure 17. Variation of π_{t-288} weight with Online DFLOD training rounds

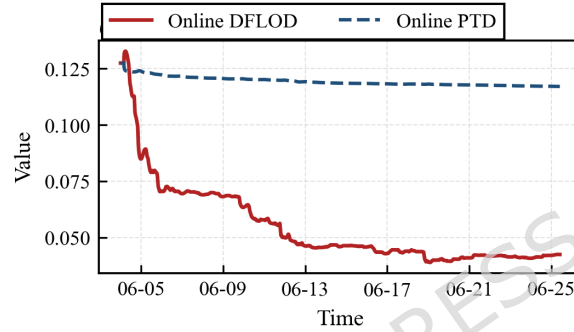


Figure 18. Variation of d_t weight with Online DFLOD training rounds

(DA-DFLOD) and intraday online execution (Online DFLOD) stages separately, and analyze the influence of key parameters such as the look-ahead window length H and the number of nodes $|\mathcal{N}|$ on online computational efficiency (Sections 5.5.1–5.5.3). The second is the component ablation and hyperparameter sensitivity analysis: ablation experiments quantify the contribution of decision-focused training, online parameter updates, delayed training, and surrogate loss smoothing to the total cost, and single-factor sweeps examine the influence of the look-ahead window, smoothing coefficient, learning rate, and SPSA perturbation radius on convergence, runtime, and final regret (Section 5.5.4). The third is the variance quantification of the SPSA zeroth-order gradient estimation and the comparison with gradient methods: at a fixed test point, we quantify the directional variance of the zeroth-order gradient estimate, give its scaling law with dimension and the control means of multi-direction averaging and learning-rate decay, and compare it with the analytic surrogate gradient based on the proxy problem and dimension-wise finite differences in terms of solve counts and runtime per step (Section 5.5.5). The fourth is the consistency analysis between the training proxy model and the execution model: the proxy objective and convex relaxation introduced in the training phase to reduce computational cost differ from the exact MIP solved in the execution phase; this section argues from two properties, monotone nesting of feasible regions and the S-stationarity guarantee, that this difference does not cause systematic bias in online parameter updates (Section 5.5.6). The fifth is the influence of SPSA randomness on the results: through multiple independent seed trials, we evaluate the influence of random perturbation directions on the online parameter update paths and final scheduling costs (Section 5.5.7). In addition, the mechanism comparison with deep reinforcement learning (DRL) control algorithms is elaborated in the discussion in Section 7, explaining the differences between the proposed DFLOD and mainstream DRL algorithms in constraint handling and dynamic update mechanisms, and their respective applicable scenarios.

5.5.1 Computational Cost Analysis of DA-DFLOD

The computational burden of day-ahead training mainly comes from the inner-level proxy scheduling solves repeatedly invoked in the SPSA zeroth-order gradient estimation. In each training round, for each sample in the training set, one base proxy scheduling and $2m$ perturbation-point solves are performed (m each for θ_k^+ and θ_k^-), totaling $1 + 2m$ proxy scheduling solves; the nominal configuration takes $m = 1$ (i.e., 3 solves per sample), and the multi-direction averaging with $m > 1$ as a variance control means is described in Section 5.5.5. It should be noted that the above counts only the proxy scheduling; in Algorithm 1, each sample per round additionally executes one execution scheduling $w_i^{*,re}$ without the smoothing term (used to record cost

and constraint violations), and the full-information benchmark $w_i^*(c_i)$ is pre-solved once per sample and is not repeated across training rounds; neither is included in the above proxy scheduling counts. If the inner-level scheduling adopts convex relaxation, it can be solved by LP or NLP with smooth nonlinear constraints. The worst-case complexity of the simplex method for LP is $O(n^3)$ (n being the number of variables), and the interior-point method is about $O(n^{2.5})$. Substituting $n = O(N_i|\mathcal{U}|^2|\mathcal{D}|)$ (N_i being the number of periods covered by the day-ahead training window), the complexity of a single proxy scheduling solve is $O((N_i|\mathcal{U}|^2|\mathcal{D}|)^{2.5})$.

SPSA requires only two function evaluations per gradient estimation (two-sided differencing), and its solve count is independent of the parameter dimension p . The total dimension of the three forecasting model parameter blocks of this paper (including the bias of each module) is $p = 14 + 1 + 11 + 1 + 30 + 1 = 58$; dimension-wise finite differences would require $2p = 116$ solves, while the nominal SPSA configuration requires only $2m = 2$ solves, significantly reducing the training burden.

DA-DFLOD uses 15 months of training samples, converges in about 200 rounds, and takes about 3 hours in total. This training is performed offline and is not included in the latency of intraday online decisions.

5.5.2 Computational Cost Analysis of Online DFLOD

The computation of Online DFLOD consists of rolling execution and delayed training. In each 15-min period, rolling execution requires feature construction, $H = 12$ -step forecasting, and one complete-window MIP scheduling solve (including the binary variables of the storage complementarity constraint); delayed training is executed asynchronously after the true information is revealed. According to Algorithm 2, each step requires $1 + 4m$ proxy scheduling solves (one base proxy, and $2m$ each for the two SPSA gradient estimations at θ_t and u_s), which is 5 solves in the nominal configuration with $m = 1$.

The MIP execution layer contains $2H|\mathcal{U}|$ binary variables, with a worst-case solve time of $O(2^{2H|\mathcal{U}|})$, but commercial solvers such as Gurobi can greatly reduce the actual runtime through cutting planes and presolving. In the configuration of this paper with 2 nodes and $H = 12$, the number of binary variables is $2 \times 12 \times 2 = 48$, and a single MIP solve is usually completed within tens of milliseconds.

To evaluate the actual online computational efficiency of Online DFLOD, a continuous online test was conducted from June 10, 2025 to July 9, 2025 (30 days in total), with 96 rolling decisions per day (2880 decisions in total), forecasting $H = 12$ steps ahead. The complete computation time of each decision (including feature construction, forecasting, scheduling solve, and delayed training) was recorded. Fig. 19 shows the distribution of the daily total computation time and the average time per decision.

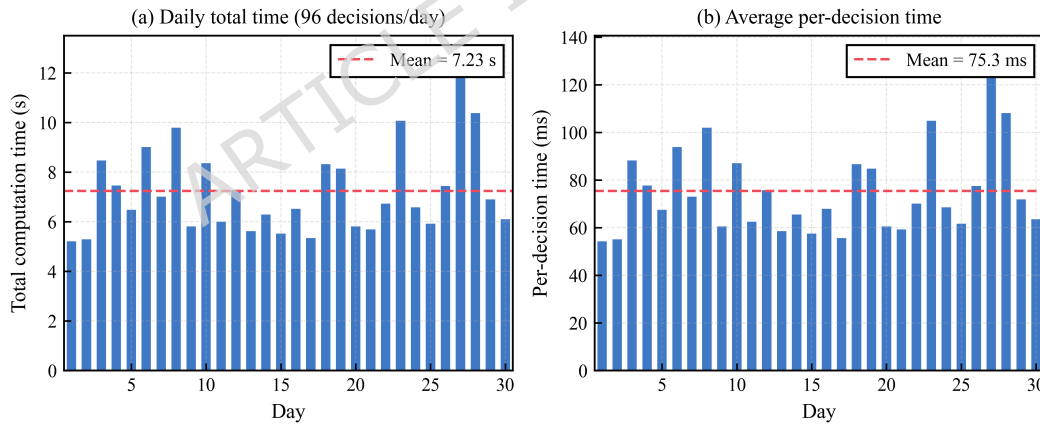


Figure 19. Online computation time statistics of Online DFLOD (June 10–July 9, 2025, 2880 decisions in total)

The cumulative computation time over the 30 days is 216.96 s, the average computation time of the 2880 rolling decisions is 75.3 ms (standard deviation 18.5 ms), with a minimum of 54.5 ms and a maximum of 133.9 ms. The per-decision time accounts for only about 0.008% of the 15-min scheduling period (900 s), fully satisfying the decision-speed requirement of real-time scheduling. The computational burden of Online DFLOD mainly comes from two parts: the Gurobi MIP scheduling solve of rolling execution (tens of milliseconds) and the proxy scheduling solves of delayed training (5 solves per step at the nominal $m = 1$, about 0.05 s in total). Delayed training is executed asynchronously after the true information is revealed and does not block the rolling decision of the current period; the forward propagation of the linear forecasting model is at the millisecond level and can be ignored. The daily total computation time fluctuates to a certain extent (5.23–12.85 s), mainly caused by the expansion of the branch-and-bound search space of the MIP solver during price spikes or task surges; nevertheless, even in the worst case (133.9 ms per solve), it is far below the 15-min scheduling limit.

Table 4 summarizes the solve counts and computational characteristics of each stage.

Table 4. Comparison of computational costs at the training stages

Item	DA-DFLOD	Online DFLOD (per step)
Proxy scheduling solves per round or step	$(1 + 2m) \mathcal{S}_u $ (nominal $m = 1$)	$1 + 4m$ (nominal $m = 1$)
Average number of perturbation directions m	1 (nominal)	1 (nominal)
Proxy scheduling type	LP or smooth NLP	LP or smooth NLP
Time per proxy solve	about 0.1–0.5 s	about 0.01 s
Gradient estimation method	SPSA	SPSA
Total computation time	about 3 h (200 rounds, 15-month samples, offline training)	about 7 s (96 rounds, one-day samples, including rolling execution and delayed training)

5.5.3 Influence of Parameters on Online DFLOD Computation Time

The look-ahead window length H is the most sensitive parameter affecting the per-step computation time of Online DFLOD, as shown in Fig. 20. H acts on three levels simultaneously: (1) the number of decision variables and constraints grows linearly with H , raising the LP proxy solve complexity as $O(H^{2.5} \sim H^3)$; (2) the number of MIP binary variables is proportional to H , and the solve-tree size grows exponentially in the worst case, but the commercial solver Gurobi (13.0.0) keeps the computation time sub-exponential through cutting planes and heuristics; (3) the per-estimation burden of SPSA is independent of H and is only affected by the increase in inner-level solve time. This paper selects $H = 12$ (3-hour look-ahead at 15-min resolution) according to the elbow rule, balancing decision sufficiency and computational burden.

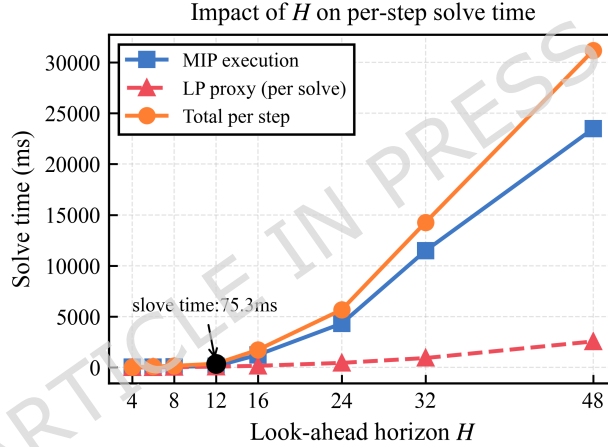


Figure 20. Influence of the Online DFLOD look-ahead window on per-step computation time

The number of data center nodes $|\mathcal{U}|$ is the second most sensitive parameter. We consider an extreme form in which all data centers are fully interconnected in pairs, used for scaling analysis. Task migration variables grow as $O(|\mathcal{U}|^2)$; therefore, when expanding from 2 nodes to 10 fully connected nodes, the number of migration variables increases from $12 \times 3 \times 2 = 72$ to $12 \times 3 \times 90 = 3240$, an increase of about 45 times. The LP proxy solve complexity increases by about $45^{2.5} \approx 1.36 \times 10^4$ times, and the number of MIP binary variables also increases from 48 to 240. When the number of fully connected data centers increases to 6, the per-step online decision and learning time is 61 s. In fact, in real data centers, computing tasks are distributed by front-end servers rather than being fully connected, and the fully connected form is used only to analyze computational cost.

The number of buffering tiers $|\mathcal{D}|$ has a linear influence on all task-related variables, and under typical values ($D = 2 \sim 3$) its influence is far smaller than that of H and $|\mathcal{U}|$. The influence of algorithm hyperparameters on the solving process and results is analyzed in the next subsection.

5.5.4 Component Ablation and Hyperparameter Sensitivity Analysis

To quantify the contribution of each major component of the algorithm and to examine the rationality of the key hyperparameter values, the component ablation experiments are conducted on the June online window and the hyperparameter sensitivity experiments on the June–December online interval. The nominal configuration refers to the parameter configuration adopted in this paper; all algorithm variants and scan points are initialized with the same day-ahead training parameters θ_{DA} to isolate the influence of the intraday online stage; in the sensitivity scan, each configuration is run with 15 independent random seeds. The component ablation results are shown in Fig. 22, and the hyperparameter sensitivity results in Fig. 23 and Fig. 24.

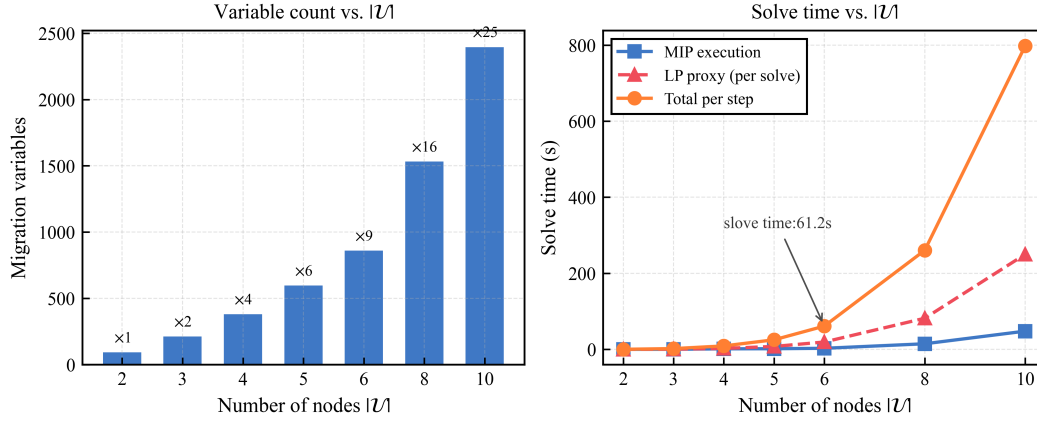


Figure 21. Influence of fully interconnected data centers on the per-step Online DFLOD computation time

The component ablation uses the total cost of the online window as the indicator (Fig. 22). The total cost of the full-information Oracle is 213991.03, and the average cost of the nominal Online DFLOD over multi-seed SPSA runs is 287488.24. After replacing the decision-focused training objective with MSE (Online PTD), the total cost rises to 420617.72, an increase of 133129.48 (46.3%) over the nominal, indicating that decision-focused training is the largest source of the cost advantage. When freezing the day-ahead parameters and retaining only rolling execution (no online parameter updates), the total cost is 309800.00, an increase of 22311.76 (7.8%) over the nominal, showing that intraday online parameter updates can further reduce the cost by about 7.2% on the basis of the day-ahead prior. When removing delayed training and using only the day-ahead training parameters (the MPC variant in Fig. 22), the total cost rises to 401522.90, an increase of 114034.66 (39.7%) over the nominal, almost reverting to the Online PTD level, indicating that inheriting the long-time-scale training result is the key mechanism of the intraday online learning gains. When the surrogate loss smoothing coefficient is $\alpha = 10^{-7}$ (almost no smoothing), the total cost is 359800.00, an increase of 72311.76 (25.2%) over the nominal, with 920 inner-level solve failures; with excessive smoothing ($\alpha = 10^{-3}$), the total cost is 369910.00, an increase of 82421.76 (28.7%) over the nominal. Overall, the four components, in descending order of cost contribution, are decision-focused training, online delayed training, surrogate loss smoothing, and online parameter updates; the surrogate smoothing coefficient needs to be moderate: too small a value loses numerical stability, and too large a value introduces surrogate bias.

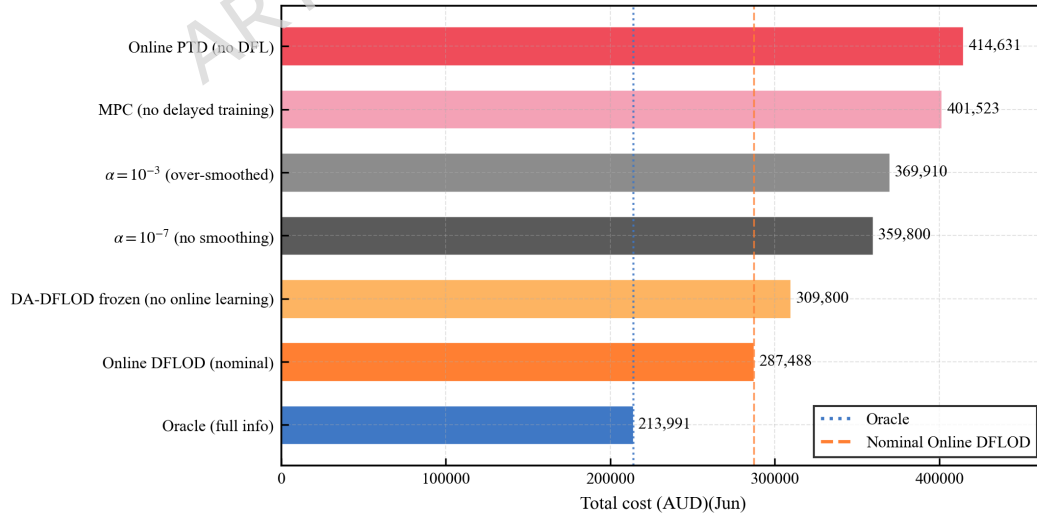


Figure 22. Component ablation results: total cost of each variant on the June 2025 online window (Oracle and nominal Online DFLOD as reference lines)

Fig. 23 presents the influence of four key hyperparameters on the final cumulative regret over the 7-month online interval test (bars, left axis) and the per-step solve time (lines, right axis). The influence of the learning rate η_1 is U-shaped: at $\eta_1 = 0.0001$,

the regret is about 2.0 times the nominal value, and the per-step time abnormally rises to about 1.31 s (caused by retries after solve failures); at $\eta_1 = 0.072$, it is about 3.3 times the nominal value; the minimum lies near the nominal value 0.018. The SPSA perturbation radius c shows large regret fluctuations at 0.02 and below, with an almost constant per-step time (about 75 ms). The influence of the smoothing coefficient α is significant: at $\alpha = 10^{-8}$ the training diverges without valid results; at $\alpha = 10^{-7}$ the regret is about 1.7 times the nominal value with the time rising to about 177 ms and constraint violations present; $\alpha = 10^{-6} \sim 10^{-5}$ forms a stable valley; at $\alpha \geq 10^{-4}$ the regret rises significantly, and the nominal 10^{-5} lies in the valley interval. When the look-ahead window H increases, the regret first decreases quickly and then saturates: the regrets at $H = 8$ and $H = 10$ are about 2.3 and 1.8 times that at $H = 12$, respectively, while $H = 16$ and $H = 24$ show little regret improvement over $H = 12$ but higher computational cost; the per-step time increases from 75.3 ms at $H = 12$ to 1150.7 ms at $H = 16$ and 5259.0 ms at $H = 24$. Therefore, $H = 12$ corresponds to the elbow of “regret saturation–computational burden,” validating the rationality of selecting $H = 12$ by the elbow rule in Section 5.5.3.

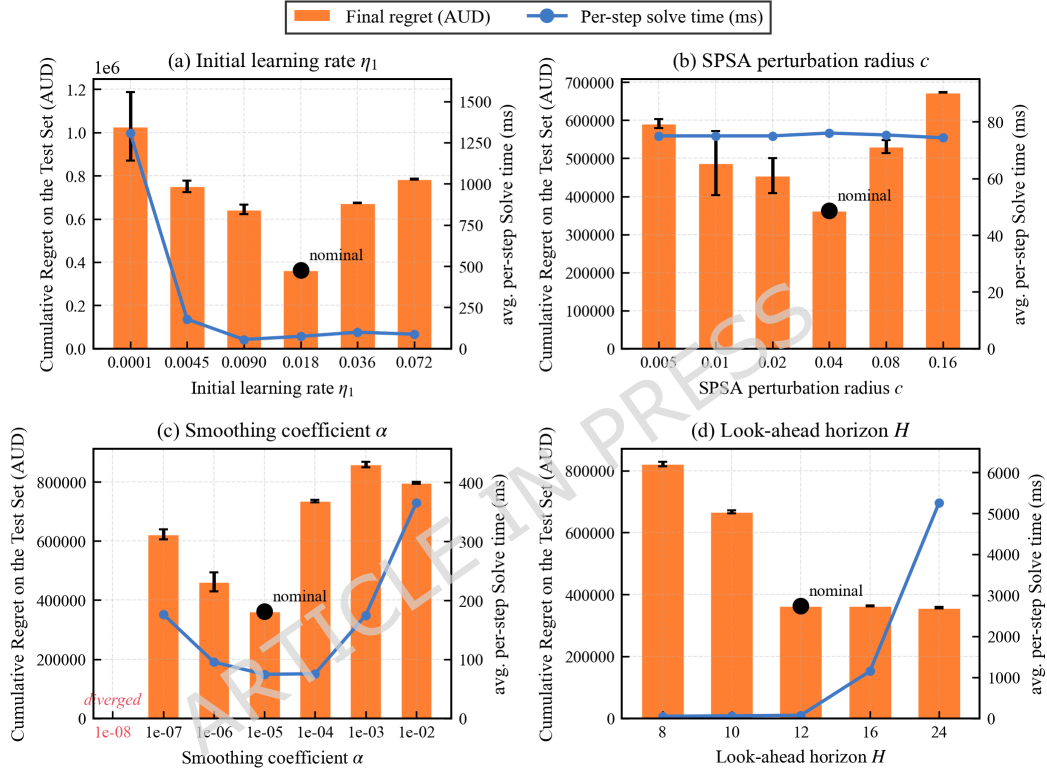


Figure 23. Hyperparameter sensitivity overview: influence of η_1 , c , α , and H on the final regret (bars, left axis) and per-step solve time (lines, right axis). Nominal values are marked with black circles; $\alpha = 10^{-8}$ diverges without valid results

Fig. 24 compares the cumulative average per-step regret (AUD/step) of June under representative values as a function of the online step number (plotted from step 200 onward, since the cumulative average per-step regret requires a certain amount of data to be computable). The nominal configurations all exhibit clear convergence behavior: the cumulative average per-step regret of $\eta_1 = 0.018$ decreases from 148.27 near step 200 to 25.2, and the $c = 0.01$ configuration oscillates but the overall trend still decreases. The degenerate configurations almost do not converge: $\eta_1 = 0.0001$ only decreases from 206.0 to 197.2, $\alpha = 10^{-7}$ decreases from 119.67 to 51.57 (accompanied by inner-level solve failures), and $\alpha = 10^{-3}$ decreases from 109.22 to 54.67, essentially leveling off. This indicates that hyperparameters determine not only the final regret but also the convergence speed and numerical stability of online training; the nominal values of this paper all lie in the interval of stable convergence.

Regarding the selection of the smoothing coefficient α_k and the gradient behavior when it is too small, this paper performs a grid scan over $\alpha \in \{10^{-8}, \dots, 10^{-2}\}$ in the online rolling test, recording the 7-month cumulative final regret, SPSA gradient norm, number of gradient-clipping triggers, number of inner-level solve failures, and storage complementarity constraint violations simultaneously. At $\alpha = 10^{-8}$, the training diverges with 1650 inner-level solve failures and no valid scheduling results; $\alpha = 10^{-7}$ can run, but exhibits 920 inner-level solve failures, a cumulative complementarity violation of 882.178, a maximum gradient norm of 14.3, and 27 gradient-clipping triggers; $\alpha = 10^{-6}$ still has 6 failures and a constraint violation of 0.0021; from $\alpha = 10^{-5}$ onward, both the failure count and constraint violations are zero, with an average gradient norm

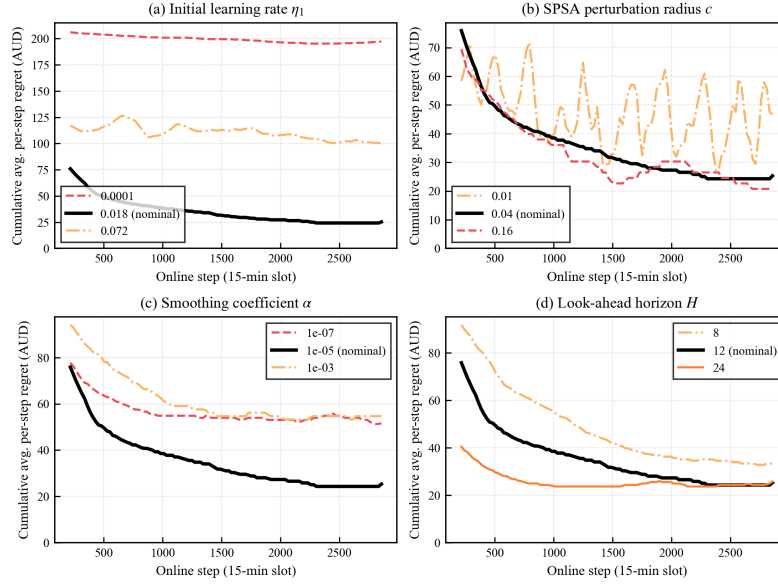


Figure 24. Cumulative average per-step regret as a function of the online step number under representative hyperparameter values (plotted from step 200 onward)

of 0.92, a maximum of 5.4, no clipping triggers, and the lowest final regret (362625.24). It should be noted that under the SPSA zeroth-order update framework of this paper, an excessively small α does not manifest as vanishing or exploding analytic gradients, but rather as noise amplification of the function-value difference near degenerate vertices, manifested as gradient-norm spikes, frequent clipping, and a surge in inner-level solve failures; after $\alpha \geq 10^{-5}$, the positive curvature provided by the quadratic smoothing term completely suppresses the above phenomena. An excessively large α introduces surrogate bias: at $\alpha = 10^{-3}$, the regret rises to 869721, about 2.4 times the nominal value. Accordingly, $\alpha = 10^{-5}$ is selected to satisfy the engineering requirement of numerical stability.

5.5.5 Variance Quantification of SPSA Zeroth-Order Gradient Estimation and Comparison with Gradient Methods

To quantitatively analyze the variance of the SPSA zeroth-order gradient estimation and compare it with other gradient methods, a diagnostic experiment is conducted at a fixed test point: look-ahead window $H = 48$, quadratic smoothing coefficient $\alpha = 1.0$ (log-barrier proxy), linear predictor parameter dimension $p = 29$, SPSA perturbation radius $c = 0.04$, uniformly using the log-barrier smoothed proxy loss, and taking the analytic surrogate gradient based on the proxy problem (implicit differentiation) as the reference.

The diagnostic experiment verifies the scaling law of the directional accuracy of the zeroth-order gradient estimate at a fixed test point, so it adopts a controlled configuration homogeneous with the main experiment. The parameter dimension takes half of the full-model $p = 58$, i.e., $p = 29$: at this dimension, the normal approximation of the single-direction cosine $\sqrt{2/\pi}/\sqrt{p} = 0.148$ is almost identical to the measured mean of 0.147, and differs from the theoretical value of 0.105 at $p = 58$ by exactly a factor of $\sqrt{2}$, providing a resolvable test of the $1/\sqrt{p}$ scaling law; meanwhile, the dimension-wise finite-difference verification ($2p = 58$ inner-level solves) is also within the diagnostic budget. The corresponding value of the full model at $p = 58$ is directly extrapolated from this validated law without additional experiments.

Over $K = 200$ independent Rademacher directions, the mean cosine similarity between the single-direction SPSA gradient estimate and the analytic surrogate gradient is 0.147 (median 0.132, standard deviation 0.115). This value is consistent with the theoretical expectation of zeroth-order estimation: the single-direction SPSA estimate can be approximately written as $\hat{g} \approx (g \cdot \Delta)\Delta$, and its cosine with the true gradient g satisfies

$$\mathbb{E}[\cos(\hat{g}, g)] = \frac{\mathbb{E}[g \cdot \Delta]}{\sqrt{p} \|g\|},$$

by Khintchine's inequality, $\mathbb{E}[g \cdot \Delta / \|g\|] \in [1/\sqrt{2}, 1]$, hence $\mathbb{E}[\cos] \in [1/\sqrt{2p}, 1/\sqrt{p}] = [0.131, 0.186]$; when $g \cdot \Delta$ is approximately normal (for large p), it is about $\sqrt{2/\pi}/\sqrt{p} = 0.148$, almost identical to the observation. Therefore, the directional cosine of the single-direction estimate decreases with dimension at the order of $1/\sqrt{p}$ (corresponding to a component variance of order $O(p)$), confirming from the mechanism the judgment that the variance of the zeroth-order gradient estimation increases

with the feature dimension (the theoretical value of the full model at $p = 58$ is about 0.105). This variance can be controlled through two approaches: (i) multi-direction averaging. After averaging the estimates of m independent directions, the cosine similarity increases from 0.15 at $m = 1$ to 0.61 at $m = 20$, 0.75 at $m = 50$, and 0.85 at $m = 100$ (Fig. 25), consistent with the variance-decreasing-as- $1/m$ law in stochastic approximation theory; (ii) adaptive learning-rate decay and parameter projection. This paper adopts a learning rate decaying with training rounds (initial value 0.018) and the parameter box-constraint projection $\Theta = [-4, 4]$, confining the per-step noise to a stable interval. The perturbation-radius scan shows that the relative deviation of the directional derivative generally increases with c (about 0.06 at $c = 0.01$, 2.4 at $c = 0.04$, and 4.8 at $c = 0.16$); the $c_k = 0.04$ of this paper strikes a balance between truncation bias and numerical stability.

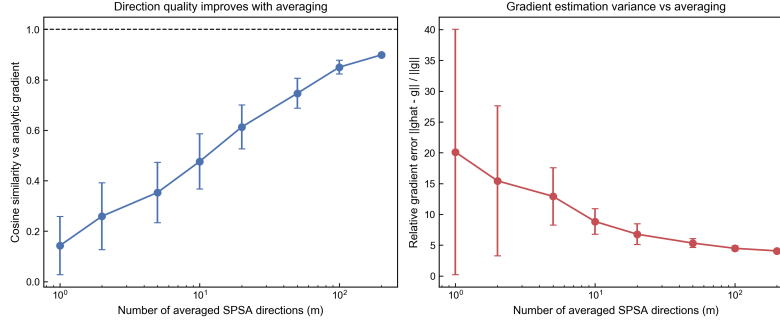


Figure 25. Influence of direction averaging on the variance of the SPSA zeroth-order gradient estimate ($p = 29$, $c = 0.04$)

To compare quantitatively with analytic gradient methods, it should first be clarified that exact KKT implicit differentiation is not directly applicable: the execution-layer MIP contains binary variables and the storage complementarity constraint (MPCC), and the lower-level problem is not differentiable everywhere—this is precisely the fundamental reason for adopting the SPSA zeroth-order estimation in this paper. As the closest analytic reference, this paper implements the analytic surrogate gradient based on the quadratic smoothing proxy problem (implicit differentiation), whose correctness is anchored by directional-derivative verification: the relative error with respect to the finite-difference directional derivative is 5.25×10^{-6} , and the relative error of the dimension-wise full gradient is 7.3×10^{-7} . Table 5 gives the per-step solve counts, runtimes, and directional accuracy of the three methods at the same test point: SPSA requires only 2 proxy solves per step (about 0.061 s, independent of the parameter dimension); the analytic surrogate gradient requires 1 solve plus the Jacobian and adjoint computation (about 0.040 s); dimension-wise finite differences require $2p = 58$ solves (about 1.51 s). It can be seen that, under the premise that the proxy is differentiable, the analytic surrogate gradient is faster per step and more accurate in direction; the value of SPSA lies in its differentiability-free requirement and the independence of the gradient-estimation solve count from the parameter dimension. The SPSA per-step gradient (2 solves) takes about 0.06 s at the $H = 48$ diagnostic point, extrapolating to about 5.9 s for 96 rounds per day; at the nominal $H = 12$, the 5 proxy solves per step of delayed training take about 0.05 s, and together with rolling execution they are consistent with the measured magnitude of about 7 s/day in Section 5.5.2; the influence of gradient-level variance on the final results is analyzed in Section 5.5.7, where the coefficient of variation of the scheduling total cost over 15 independent seeds is only 0.055%, indicating that the instantaneous noise of the single-direction estimate after iterative averaging does not change the economic conclusions.

Table 5. Comparison of gradient estimation methods (fixed test point, $H = 48$, $p = 29$)

Item	SPSA	Analytic surrogate gradient (implicit differentiation)	Dimension-wise finite differences
Inner-level solves per step	2	1 (including Jacobian/adjoint)	$2p = 58$
Time per step (s)	0.061	0.040	1.51
Relation to parameter dimension	Independent	Requires Jacobian computation	Linear
Directional accuracy	Mean cosine 0.147	Directional relative error 5.25×10^{-6}	Full-gradient relative error 7.3×10^{-7}
Applicable condition	No differentiability required	Differentiable proxy layer	Small parameter dimension

5.5.6 Consistency Analysis Between the Training Proxy Model and the Execution Model

When the sufficient conditions for the optimal convex relaxation do not hold, the training stages of Sections 4.1.3 and 4.2.2 adopt the quadratic smoothing proxy objective (Equation (22)) and the convex relaxation or regularized feasible region (Equations (9) and (21)), while the actual execution in Section 5.1 adopts the exact complementarity-constrained MIP solve with binary variables. The question is whether the execution layer can remain consistent with the MIP solution if the regularization method is also used. This section shows that this difference does not cause systematic bias in online parameter updates.

The essence of the above difference is to use a regularized approximation of the MPCC for parameter updates. Reference⁴⁴ establishes a general theoretical framework for MPCC regularization. In the following, combined with the storage complementarity constraint of this paper, two core properties and their brief proof ideas are given.

Property 1 (Monotone nesting of feasible regions). Let the storage parts of the original complementarity feasible region and the regularized feasible region be $\mathcal{W}^s$ and $\mathcal{W}^{s,\text{reg}}(\varepsilon)$, respectively. For any $0 \leq \varepsilon_1 \leq \varepsilon_2$, $\mathcal{W}^{s,\text{reg}}(\varepsilon_1) \subseteq \mathcal{W}^{s,\text{reg}}(\varepsilon_2)$, and $\lim_{\varepsilon \rightarrow 0^+} \mathcal{W}^{s,\text{reg}}(\varepsilon) = \mathcal{W}^s$. Proof idea: when $w_t^c + w_t^d \geq 2\varepsilon_2$, if $(w_t^c - \varepsilon_2)(w_t^d - \varepsilon_2) > 0$, the two factors have the same sign; both being negative contradicts $w_t^c + w_t^d \geq 2\varepsilon_2$, and both being positive implies by $\varepsilon_1 \leq \varepsilon_2$ that $(w_t^c - \varepsilon_1)(w_t^d - \varepsilon_1) > 0$, contradicting $\Phi_t(\cdot, \varepsilon_1) \leq 0$. When $w_t^c + w_t^d < 2\varepsilon_2$, $\Phi_t(\cdot, \varepsilon_2) \leq 0$ holds automatically. Therefore, the regularized feasible region shrinks monotonically to the original feasible region as $\varepsilon \rightarrow 0^+$, without introducing points infeasible in the original problem.

Property 2 (S-stationarity guarantee). Let $\{\varepsilon_k\} \downarrow 0$ and $\{x^k\}$ be a sequence of KKT points of the storage-constraint relaxed problem with $x^k \rightarrow x^*$. If the CPLD constraint qualification holds at x^* , then x^* is an S-stationarity point of the original MPCC—that is, the strongest KKT-type first-order necessary condition for MPCCs. Proof idea, in three steps: (i) the boundedness of the multiplier sequence is proved by CPLD: if the multiplier norm diverges, taking the limit direction yields positive linear dependence of the constraint gradients, contradicting the validity of CPLD; (ii) taking the multiplier limit and reorganizing yields a W-stationarity multiplier of the original MPCC; (iii) using the structure of the complementarity index set, excluding the “W-but-not-M” cases yields M-stationarity, which is then strengthened to S-stationarity for the special structure of the storage complementarity constraint. Reference⁴⁴ gives the standard convergence path of M-stationarity.

Properties 1 and 2 jointly show that the stationary points of the regularized proxy problem are exactly the true stationary points of the original problem, that the regularization itself does not introduce spurious stationary points, and that the optimal solutions of the proxy layer can arbitrarily approximate feasible solutions within the original feasible region when ε is sufficiently small. Under this theoretical guarantee, the gradient descent direction given by the training proxy model is, in a probabilistic sense, a correct descent direction of the true decision loss. Meanwhile, the SPSA zeroth-order gradient estimation relies only on function-value comparisons rather than exact gradients, and is naturally robust to the numerical residual between the proxy layer and the execution layer; the quadratic smoothing coefficient α takes a sufficiently small value in training (1.0×10^{-5} , see Table 2), and its perturbation to the optimal-solution location is negligible. The experimental results of Section 5.2 also confirm this: Online DFLOD has zero storage constraint violations in all test months, always outperforms Online PTD in operating cost, and stays close to the Oracle lower bound, indicating that the proxy training does not cause systematic suboptimality bias.

5.5.7 Influence of SPSA Randomness on the Results

The SPSA zeroth-order gradient estimation approximates the decision-loss gradient by randomly sampling perturbation directions in the parameter space; the random direction sequence differs in each training run, which may cause differences in the online parameter update path and final scheduling cost of Online DFLOD. To evaluate the influence of this randomness on the robustness of the conclusions, the complete intraday rolling scheduling procedures of Online DFLOD, Online DFLOD-RN, and Online PTD are independently run with 15 different random seeds under the same data partition and hyperparameter settings. The Oracle benchmark is not affected by SPSA randomness and is run only once. Table 6 summarizes the mean, standard deviation, and 95% confidence interval of the total scheduling cost of the three methods over the 15 independent trials. The confidence intervals are computed based on the t -distribution, which is used to estimate confidence intervals for small samples (degrees of freedom $n - 1 = 14$, two-sided $t_{0.025,14} = 2.145$).

Table 6. Scheduling cost statistics under different random seeds ($n = 15$)

Method	Mean total cost	Std. dev. of total cost	95% CI of total cost
Online DFLOD	287488.24	157.47	(287401.00, 287575.50)
Online DFLOD-RN	263438.91	141.01	(263360.81, 263517.01)
Online PTD	420617.72	140.34	(420539.99, 420695.45)

As can be seen from Table 6, the mean total cost of Online DFLOD over the 15 independent trials is 287488.24, and that of Online PTD is 420617.72. The coefficient of variation (CV) of the cost is 0.055% for Online DFLOD and 0.034% for Online PTD, both at low levels, indicating that the variation of the SPSA random perturbation directions has little influence on the cost consistency of the two methods, and the methods have stable repeatability. The 95% confidence intervals of the two methods do not overlap (the upper bound of Online DFLOD, 287575.50, is far below the lower bound of Online PTD, 420539.99), indicating that the economic advantage of Online DFLOD over Online PTD is statistically significant and not caused by chance in SPSA random sampling. Online DFLOD-RN, with the nonlinear predictor, improves the cost by about 8% over Online DFLOD with the linear predictor; its mean over the 15 trials is 263438.91 with a CV of 0.054%, and its 95% confidence interval (263360.81, 263517.01) does not overlap with that of the linear Online DFLOD, showing that the cost improvement is equally stable across different random seeds.

The variance of the SPSA gradient estimation mainly comes from two sources: one is the sampling variance of the Rademacher random perturbation directions: each step requires only two function evaluations and the solve count is independent of the parameter dimension, but the directional accuracy decreases with dimension at the order of $1/\sqrt{p}$ (see Section 5.5.5), and its estimation efficiency has been discussed in Section 4.1.4; the other is the randomness of the true electricity prices and task arrivals in the intraday rolling window, which affects Online DFLOD and Online PTD identically and does not amplify the relative variability between the two methods.

In addition, the storage complementarity constraint violation ($\max w_{t,u}^c, w_{t,u}^d$) is zero under all seeds, and the SOC bound violations and solver failure counts are also zero, indicating that the regularized proxy training method stably maintains physical feasibility under different random seeds.

In summary, the performance advantage of Online DFLOD is not significantly affected by the SPSA random perturbation directions, and the cost saving is statistically significant.

6 Conclusion

This paper addresses the problem of multiple interconnected data centers with distributed PV, energy storage, and task migration constraints. By constructing an Online DFLOD framework that integrates forecasts of electricity prices, PV generation, and task arrivals, the forecasting model is trained with scheduling cost rather than mere forecast error as the training objective. A two-stage mechanism combining day-ahead training with intraday rolling execution and delayed training is proposed, enabling the model to continuously correct short-term deviations based on long-term features and revealed operational data. A linear model of task backlog and spatiotemporal transfer suitable for DFL is established, allowing cross-center task migration, queue state propagation, and SLA constraints to participate in training as the lower-level optimization layer. The non-differentiability and numerical oscillation issues arising from the storage complementarity constraint are mitigated through a regularized surrogate loss. Case study results show that the proposed Online DFLOD method outperforms conventional online predict-then-decide methods in every month from June to December 2025, reducing monthly energy costs by 16.5%–32.4% with zero storage constraint violations, and increasing the PV consumption rate from 64.1% to 78.3%.

This paper focuses on establishing the decision-focused online learning framework and validating its core mechanism; the applicability boundaries of the above modeling simplifications, the limitations of the method, and future research directions are systematically elaborated in the discussion in Section 7.

7 Discussion

This paper proposes a decision-focused online learning framework with day-ahead training and intraday rolling scheduling, which makes the forecasting models of electricity prices, PV output, and task arrivals directly train toward the scheduling cost, and adapts to the operational form of gradually revealed information through rolling execution and delayed updates. When constructing the lower-level scheduling model, in order to highlight and validate the effectiveness of the decision-focused online learning mechanism itself while taking solving efficiency into account, this paper adopts simplified modeling of the cooling system, communication network, task queue, storage degradation, and renewable output consistent with the 15-min scheduling granularity. The basis, applicability boundaries, and influence on the conclusions of these assumptions are discussed item by item below, and the research limitations and future directions are summarized accordingly.

First, we discuss the thermodynamic simplification on the energy-consumption side of the data center. In Equation (3.3.2), the total power consumption of the data center is converted from the IT equipment power consumption through a static PUE coefficient, without explicitly characterizing the dynamic processes of chillers, air handling units, and building thermal inertia. This simplification is consistent with the 15-min scheduling period scale of this paper: transient thermal experiments and CFD studies point out that the thermal time constants of the main thermal masses such as servers are only on the order of several minutes⁴⁶; after accounting for the thermal capacity of servers and racks, the time for the system to reach 95% of the new steady state is about 500 s (about 9 min)⁴⁷, both smaller than the scheduling period of this paper. Therefore, within each period, the IT heat generation and cooling load can be approximately regarded as being in quasi-steady state; the dynamic energy consumption of chillers and air handling units does not transfer significantly across periods with the spatiotemporal migration of tasks, and cooling constraints such as return-air temperature and cooling capacity reduce, in a period-averaged sense, to constraints on the total power level of each node in the period, which, together with the processing capacity constraint (13f) after PUE conversion, jointly limit the acceptable task volume of each node, thereby influencing the task scheduling feasible region and the energy-cost optimization results. It should be noted that the above conclusion is applicable under the 15-min scheduling granularity and conventional air-cooled machine rooms; in scenarios with a shorter scheduling cycle or significantly higher room thermal density, explicit dynamic cooling modeling remains a valuable research direction.

Second, we discuss the simplification of inter-node communication and data transmission. The model abstracts inter-node communication as a link bandwidth constraint (13b), ignoring the influence of network-layer factors such as communication

latency, data loss, network security, and data transmission cost on scheduling. This simplification can be supported by measured data of production cross-data-center wide-area networks: the maximum round-trip latency of the entire network of Google's private WAN B4 deployed for global data centers is only about 150 ms⁴⁸, nearly four orders of magnitude smaller than the scheduling period; Microsoft's software-driven wide area network SWAN (SWAN; a research prototype reported in⁴⁹, <https://doi.org/10.1145/2486001.2486012>) deployed for production interconnected data centers aggregates the traffic demands of each node in 5-min cycles and synchronizes the traffic engineering configuration of the entire network, with a single network configuration update requiring only 1–3 steps⁴⁹, also far faster than the scheduling period. Therefore, at the 15-min granularity, the control information required for task migration and the state information of task arrivals, electricity prices, and PV output can be synchronized within the scheduling time, and communication latency does not change the period-level scheduling decision; centralized traffic management limited by link bandwidth is precisely the prevailing operation mode of the above production-grade interconnected data center WANs. In terms of cost, since the interconnected data centers considered in this paper belong to the same computing service operator, the migration traffic between nodes is carried by the operator's own backbone network without incurring market-based public network usage charges, and the link bandwidth can be regarded as an already-invested fixed resource whose construction and operation costs do not change with a single scheduling decision; hence, the objective function does not explicitly include migration cost. In addition, the state information transmitted between nodes (queue backlogs, SLA states, prediction sequences, etc.) is small in volume, and occasional data loss is corrected by the online rolling execution and delayed training mechanisms with measured data after the true information is revealed; the SLA delay margin of $D = 2$ (30 min) also provides additional fault tolerance. Communication security is guaranteed by the operator through authentication and encryption mechanisms on dedicated links, and the protocol overhead is negligible relative to the migration data volume and has been included in the link bandwidth margin. When inter-node public bandwidth is leased, cross-operator usage-based billing exists, or network transmission energy consumption and delay penalties need to be explicitly evaluated, the framework of this paper can be naturally extended to add a communication cost term linearly related to the migration volume in the objective function. The protection against security attacks such as malicious data injection is orthogonal to the scheduling optimization of this paper; engineering issues such as stress tests under abnormal events, communication robustness, data quality, and security mechanisms will be left for future work to verify in combination with specific hardware platforms and communication protocols.

Third, we discuss the simplification of task modeling. Homogeneous tasks and FIFO queues are prevailing assumptions for aggregate load modeling of data centers^{3,50–52}. At the 15-min scheduling granularity, the energy cost depends only on the total processing volume and its reallocation across time and space; individual task differences are averaged out at the aggregate scale, and the model only needs to retain delay tolerance as the degree of freedom, which is consistent with the delay-tolerant batch-task-dominated scenario of this paper; the delay tier set $\{0, \dots, D\}$ is a model parameter, and $D = 2$ (30 min) is the calibrated value for the SLA of the studied cluster. The extension to heterogeneous tasks does not change the linear structure of state transitions: it suffices to replicate the queue variables of Equations (12a)–(12f) by class and set each delay tier independently; delay-sensitive tasks can be characterized as rigid loads by reducing the delay tier, at the cost of an increased state dimension and more complicated attribution of training signals across classes. DAG-type tasks require the introduction of integer decisions, which is not considered in this paper; approximate training stability under preemptive execution and DAG tasks is left for future work.

Fourth, we discuss the simplification of storage degradation modeling. Equation (7a) approximates storage degradation with a linear cycling cost $\gamma^{\text{ess}}(w_{t,u}^c + w_{t,u}^d)$, whose role is to make the scheduling model suppress meaningless charging/discharging actions under small price spreads with a cost proportional to the charging/discharging throughput, and to reflect the differences in storage usage intensity across scheduling strategies. The research objective of this paper is to validate the effectiveness of the decision-focused online learning mechanism itself: under fixed storage equipment parameters and scheduling granularity, all compared methods share the same degradation cost basis, and the storage-usage-intensity differences reflected by the linear approximation are already sufficient to support mechanism-level conclusions. More refined storage degradation models—such as nonlinear degradation models considering depth of discharge, state of charge, temperature, and C-rate, or capacity-loss accounting based on annual SOH decay—are mainly used to precisely quantify storage lifetime and operating revenue, and are more important for fine-grained storage economics investment calculation and revenue assessment, where the accuracy of the degradation model directly affects the long-term evaluation results. Therefore, this paper adopts the linear approximation under the premise of focusing on mechanism validation, and lists fine-grained storage degradation modeling as a future research direction.

Fifth, we discuss the predictor representation capacity of the forecasting model. This paper adopts the linear predictor as the main experimental model, to highlight the decision-focused online learning mechanism itself while keeping the prediction parameters interpretable, so that the prediction-parameter trajectories directly correspond to feature importance (Section 5.4). Section 3.4 further designs a ResNet nonlinear predictor with residual connections; the comparative experiments in Sections 5.2 and 5.5.7 show that the proposed framework is general across predictors with different representation capacities: Online

DFLOD-RN reduces the cost by about 8.2% over Online DFLOD while its MSE only decreases slightly, indicating that the benefit of the nonlinear predictor mainly comes from the adjustment of the prediction–scheduling mapping relationship by decision-focused training rather than a pure forecast-accuracy improvement; this also suggests that, under the current task scale and feature set, the linear predictor has already captured most of the information critical to scheduling decisions. Designing dedicated predictor structures for the respective time-series characteristics of electricity prices, PV output, and task arrivals, and systematically exploring the performance improvement space of decision-focused learning under stronger representation capacity, are directions worth pursuing in depth.

Sixth, we discuss the simplification of renewable output modeling. This paper considers only distributed PV, uses a unified linear predictor for renewable forecasting, and takes the radiation flux and temperature channels at the grid points of the PV stations of each node as the NWP inputs (Section 3.4); in the scenario of this paper, the installed PV capacity of each node is small, and the center-grid-point features can already provide the cloud-cover and irradiance information needed for forecasting, so this simplification does not affect the validation of the decision-focused online learning mechanism. As data centers incorporate wind power or larger-scale wind–solar complementary generation, the fluctuation characteristics, forecasting difficulty, and curtailment cost structure of renewable output will all change; wind–solar hybrid energy modeling will be included in future work as part of the physical system modeling refinement.

Seventh, we discuss the applicability of the method on larger interconnected networks. The computational cost analysis in Section 5.5.3 shows that the cost of centralized solving rises rapidly with network scale: the per-step solve time is most sensitive to the look-ahead window H and the number of nodes $|\mathcal{N}|$; when H increases from 12 to 24, the per-step time rises from 75.3 ms to 5259.0 ms, and when the number of nodes expands from 2 to 10 (fully connected form), the number of task migration variables grows by about 45 times, with the LP proxy solve complexity increasing by approximately four orders of magnitude accordingly. It should be noted that the fully connected form is used only for computational cost analysis; in reality, task distribution in data centers is usually completed by front-end servers, and the coupling of the actual topology is lower than the fully connected assumption. Even so, for larger-scale interconnected data centers, the scalability of centralized solving and centralized training will still constitute a bottleneck, which also provides research space for distributed decision-focused learning methods.

Finally, we discuss the relationship between the proposed method and deep reinforcement learning control algorithms. Deep reinforcement learning (DRL) has become a mainstream solving paradigm for continuous control problems, of which PPO⁵³, SAC⁵⁴, and MBPO⁵⁵ are the most widely used, and mature means such as reward shaping, Lagrangian duality, action repair, and safe policy optimization have been developed for constraint handling. The DFLOD of this paper chooses another implementation path: all operational constraints are explicitly retained in the lower-level optimization layer; the training stage uses convex relaxation and the quadratic smoothing proxy to guarantee the computability of the decision-loss gradient, and the execution stage directly solves the exact MIP containing the complementarity constraint, so that constraint satisfaction is guaranteed by the optimization solve in the form of feasible solutions; Properties 1 and 2 in Section 5.5.6 further guarantee the consistency between the proxy layer and the execution layer. From the dynamic update dimension, the intraday mechanism of DFLOD takes the decision regret rather than the prediction error or reward signal as the update objective, with minimal per-step update overhead that does not block decisions; however, this design presupposes that the optimization layer can be embedded in training, and has its own applicable scenarios compared with the “sample-update” paradigm of DRL. It should be frankly stated that after incorporating the model of this paper (including the MPCC constraint and the data center cycle constraint) into a 15-month, 200-round DRL day-ahead training framework, the action-repair results of the decisions never simultaneously satisfy the storage and data center task transfer constraints, possibly for the following reasons: first, the storage complementarity constraint and the task spatiotemporal transfer constraint are deeply coupled across periods—the former requires period-by-period feasibility of charging/discharging complementarity, while the latter requires cross-period consistency of queue backlogs, cross-center migration, and the cycle constraint—and repair in a high-dimensional action space usually adopts period-by-period or module-by-module local projections, which makes it difficult to coordinate the two types of constraints in a single repair, and repairing one often destroys the other; second, the geometric structures of the feasible regions of the two types of constraints differ greatly, the joint projection lacks an analytic closed-form solution, and alternating iterative repair cannot stably converge to a simultaneously feasible point within a limited number of steps; third, the repair order and step size are sensitive to the repair results, and under the large-scale samples of 15-month, 200-round day-ahead training, it is difficult to obtain a consistent and reproducible repair strategy within a limited tuning budget. This work is therefore left for subsequent dedicated research. When the action space lacks an analyzable optimization structure, is difficult to characterize with an optimization model, or allows soft constraints in exchange for control flexibility, DRL remains a suitable alternative.

In summary, the method proposed in this paper still has limitations in the following aspects: physical system links such as storage degradation, cooling energy consumption, task queues, and renewable types are simplified consistently with the scheduling granularity; engineering issues such as communication robustness, data quality, and security mechanisms rely on indirect support from measured production WAN data and have not yet been validated on site in combination with specific

hardware platforms and communication protocols; the scalability of centralized solving and centralized training on larger networks remains to be examined; and numerical comparison with mainstream DRL algorithms has not yet been completed, mainly due to the insufficient effective handling of the MPCC constraints and the large-scale day-ahead training set at the implementation level, and the related comparison is left for subsequent dedicated research.

Based on the above discussion, future work will proceed along the following directions.

First, the combination of predictor representation capacity and decision-focused training. This paper adopts the linear predictor as the main experimental model to highlight the decision-focused online learning mechanism itself while keeping the prediction parameters interpretable; the comparative experiments with the nonlinear predictor show that the proposed framework is general across predictors with different representation capacities. In the future, dedicated predictor structures can be designed for the different time-series characteristics of electricity prices, PV output, and task arrivals, systematically exploring the performance improvement space of decision-focused learning under stronger representation capacity.

Second, refinement of physical system modeling. This paper adopts simplifications consistent with the scheduling granularity for links such as storage degradation, cooling energy consumption, task queues, and renewable types. In the future, while maintaining a lower-level optimization structure embeddable in DFL training, more refined storage degradation, dynamic cooling, heterogeneous task, and wind-solar hybrid energy models can be gradually introduced, and the influence of the increased model complexity on training stability and online computational efficiency can be evaluated.

Third, engineering validation combined with production environments. Engineering validation oriented toward actual deployment, including stress tests under abnormal events, communication robustness, data quality, and security mechanisms, will be pursued as future applied research in combination with specific hardware platforms and communication protocols.

Fourth, extension to large-scale interconnected networks. For larger-scale interconnected data centers, the cost of centralized solving rises with network scale, which also provides research space for distributed decision-focused learning methods. In the future, distributed decision-focused learning methods based on decomposition-coordination or hierarchical control can be studied to extend the proposed framework to large-scale networks.

Fifth, systematic comparison with advanced learning methods. Systematically comparing the proposed method with mainstream deep reinforcement learning algorithms such as PPO, SAC and MBPO as well as more advanced deep learning forecasting models at the numerical level is important work for further validating the practical value of the framework. Such a comparison needs to be rigorously completed under exactly the same rolling execution, constraint handling, and statistical validation basis as this paper, and constitutes a structurally complete independent study; we have been advancing it as a follow-up dedicated work and plan to report the related results in the form of a full paper.

Data availability

Upon reasonable request, the corresponding author will provide the data supporting the study's conclusions.

Code availability

The code used in this study is openly available at <https://doi.org/10.5281/zenodo.21927607> (DOI: 10.5281/zenodo.21927607).

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Competing interests

The authors declare no competing interests.